

A Dimension-Free Bound on the Poincaré Constant of Isotropic Log-Concave Measures

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Abstract

The Kannan–Lovász–Simonovits (KLS) conjecture asks whether isotropic log-concave probability measures satisfy a Poincaré inequality with a constant independent of dimension. In this paper, we establish a dimension-free Poincaré inequality and hence a universal positive lower bound for the Euclidean Cheeger constant, thereby proving the KLS conjecture. Our proof has three steps. First, we study integration operators that undo differentiation and prove a curvature estimate that is uniform in the number of tensor indices. Second, an operator lemma turns bounds for low-degree polynomials into bounds for any number of integrations, with one common multiplicative factor. Third, we use stochastic localization to transfer the integration bounds back to polynomial norms.

1 Introduction

The Kannan–Lovász–Simonovits (KLS) conjecture asks whether the Poincaré constant of a log-concave measure is bounded by a universal multiple of its largest covariance eigenvalue (Kannan et al., 1995). The conjecture thus predicts that one-dimensional variances control high-dimensional isoperimetry and concentration, with constants independent of dimension (Milman, 2009). Its consequences include thin-shell bounds, quantitative central limit results for convex bodies, and improved guarantees for sampling, see, e.g., (Lee and Vempala, 2024; Chen and Eldan, 2025).

Let $n \geq 1$ be an integer and let μ be a probability measure on $\mathbb{R}^n$. Its affine support is the smallest affine subspace carrying the measure. The measure is log-concave if its density on this subspace has the form e^{-V} , where V is convex and takes values in $\mathbb{R} \cup \{+\infty\}$. Uniform measures on convex bodies and Gaussian measures are basic examples. A measure on $\mathbb{R}^n$ is isotropic if its mean is zero and its covariance is Id , the $n \times n$ identity matrix.

For a probability measure μ on $\mathbb{R}^n$, let $C_P(\mu) \in [0, \infty]$ denote the least constant in

$$\text{Var}_\mu f \leq C_P(\mu) \int |\nabla f|^2 d\mu.$$

We initially test this inequality on smooth real-valued functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Appendix F extends it to locally Lipschitz functions whose squared gradient is integrable. If the measure lies in a

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proper affine subspace, gradients are taken within that subspace. Linear functions show that $C_P(\mu) \geq 1$ in isotropic position. The conjecture asserts that $C_P(\mu)$ is also bounded above by a universal constant, independently of the dimension.

We compare the known bounds using the Cheeger constant $h(\mu) \in [0, \infty]$ and its worst-case reciprocal $\psi_n \in [0, \infty]$. They are defined by

$$h(\mu) = \inf_{0 < \mu(E) < 1} \frac{\text{Per}_\mu(E)}{\min\{\mu(E), 1 - \mu(E)\}}, \quad \psi_n = \sup_\mu h(\mu)^{-1},$$

where $E \subseteq \mathbb{R}^n$ is Borel and the supremum is over isotropic log-concave measures on $\mathbb{R}^n$. The numerator measures the boundary of a set; the denominator is the smaller of its mass and that of its complement. We use the outer Minkowski boundary measure

$$\text{Per}_\mu(E) = \liminf_{\varepsilon \downarrow 0} \frac{\mu(E_\varepsilon) - \mu(E)}{\varepsilon}.$$

Here, $E_\varepsilon = \{x \in \mathbb{R}^n : \text{dist}(x, E) < \varepsilon\}$ is the open ε -neighborhood of E , for $\varepsilon > 0$. For a measure supported on a proper affine subspace, neighborhoods and distances are taken within that subspace.

For log-concave measures, $C_P(\mu)$ and $h(\mu)^{-2}$ are comparable up to universal constants. Ledoux proved the implication from a Poincaré bound to a Cheeger bound under smooth convexity assumptions; see Milman (2009, Theorem 1.4). Milman's equivalence between isoperimetry, spectral gap, and concentration (Milman, 2009, Theorem 1.5) also covers arbitrary absolutely continuous log-concave probability measures on $\mathbb{R}^n$. Thus, the conjecture is equivalent to $\sup_{n \geq 1} \psi_n < \infty$. All historical bounds below concern ψ_n ; the corresponding Poincaré bounds have the square of this dependence. We state these rates for $n \geq 3$, with C denoting a positive universal constant.

Kannan et al. (1995) obtained the original $C\sqrt{n}$ bound for isotropic convex bodies by localization. The same argument extends to log-concave densities, giving $\psi_n \leq C\sqrt{n}$; see Chen (2021, Section 1).

The stochastic localization method introduced by Eldan (2013) evolves a measure by multiplying its density by Gaussian factors with randomly changing centers. Estimates for the evolving covariance then give functional inequalities for the initial measure. This method related concentration of the Euclidean norm to the spectral gap, up to logarithmic factors, and led to a sequence of improvements. In work first published in 2017, Lee and Vempala (2024) proved $\psi_n \leq Cn^{1/4}$. Chen (2021) then established the subpolynomial estimate

$$\psi_n \leq \exp(C\sqrt{\log n \log \log n}), \quad n \geq 3.$$

Klartag and Lehec (2022) obtained the first polylogarithmic bound, $\psi_n \leq C(\log n)^5$. Jambulapati, Lee, and Vempala (2022) reduced the exponent to 3.2226. Klartag (2023) subsequently proved $\psi_n \leq C\sqrt{\log n}$, using an improved curvature estimate for the Poincaré constant.

More recently, Letwin (2026, Theorem 1.1) proved $\psi_n \leq C(\log n)^{1/4}$. His principal input is the dimension-free variance bound for quadratic forms in Equation (2.8). It controls the third moments that drive covariance fluctuations in stochastic localization.

During the final stages of preparation of this manuscript, a recent work by Song and Zhang (2026, Theorem 7.1) combined higher-degree polynomial estimates with an iteration of curvature bounds to obtain

$$\psi_n \leq C 4^{\log^*(n+2)}.$$

Here, $\log^* x$ is the least number of successive natural logarithms needed to reduce x to at most one. Their estimate places the remaining dimension dependence in a factor that accumulates over successive iterations.

Our proof closes a direct induction on polynomial degree using bounds for repeated integration with a common prefactor. Covariance control during localization comes from the quadratic variance estimate of Letwin (2026, Theorem 1.2, Equation (1.2)), stated below as Equation (2.8).

Theorem 1.1 (Uniform Poincaré Bound). *There is a universal constant $K_P > 0$ such that every log-concave probability measure μ with $\text{Cov}(\mu) \preceq \text{Id}$ satisfies*

$$\text{Var}_\mu f \leq K_P \int |\nabla f|^2 d\mu$$

for every real-valued locally Lipschitz function f on the affine support of μ with $\int |\nabla f|^2 d\mu < \infty$. Gradients are intrinsic to the affine support.

An explicit admissible value of K_P is given in Appendix G.

Corollary 1.2 (Uniform Cheeger Bound). *There is a universal constant $K_{\text{Ch}} > 0$ with the following property. Every isotropic log-concave probability measure μ of positive dimension satisfies*

$$h(\mu)^{-1} \leq K_{\text{Ch}},$$

with Euclidean distances and perimeter taken within the affine support.

The corollary follows from Theorem 1.1 by the Poincaré–Cheeger comparison for log-concave measures (Milman, 2009, Theorems 1.4 and 1.5).

Relation to prior work. Our argument builds on a long line of work on stochastic localization and the KLS conjecture. We use the covariance-normalized localization process of Eldan (2013, Section 2), within the framework of Chen and Eldan (2025, Section 2.4.2). Our polynomial estimates use the classical Appell family (Appell, 1880; Anshelevich, 2004). We use the same normalization as Song and Zhang (2026, Section 2.3). Letwin’s quadratic inequality controls covariance fluctuations and supplies the degree-two coefficient bound used to start the induction. The Hodge comparison, the common prefactor for all integration powers, and the transfer using several derivatives are the three steps developed here.

1.1 Proof Overview

The proof has three main steps:

(1) Controlling a single integration: The derivatives of a scalar function are symmetric tensor fields whose next derivatives remain fully symmetric. We call fields with this property *compatible*. For a smooth, positively curved measure, every square-integrable compatible field of positive rank has a unique antiderivative with mean zero. Section 4 constructs the resulting integration operators. The weighted divergence used to estimate these operators can destroy compatibility. Our Hodge estimate bounds the loss from projecting its output back onto compatible fields, retaining the same curvature lower bound at every tensor rank (Lemma 3.1). Its proof is in Appendix A; the domain and closure arguments are in Appendix B.

(2) Controlling repeated integrations: Taking d centered primitives of a constant tensor of rank $d \geq 1$ gives an Appell polynomial divided by $d!$. The operator lemma in Section 5 combines bounds for these constant inputs through a finite degree with the curvature comparison. It controls any number of integrations on arbitrary compatible inputs with mean zero, provided their rank

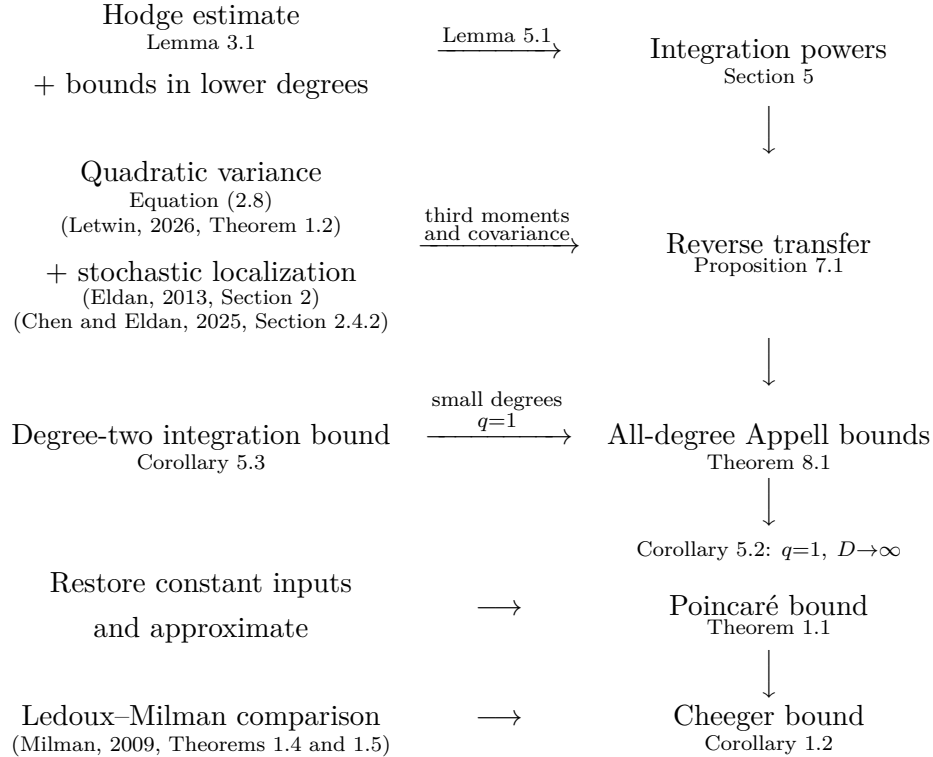
allows the integrations. The same multiplicative prefactor works for every power. This is stronger than multiplying separate bounds for each integration. Appendix C proves the operator lemma.

(3) Bounding polynomials by induction: Stochastic localization introduces positive curvature while Letwin’s quadratic variance inequality controls covariance fluctuations (Section 6). We follow the degree- d Appell polynomial of the changing measure, keeping its leading tensor fixed. Its derivatives of orders below d have mean zero. For $1 \leq q \leq d - 1$, we recover the polynomial by integrating its q th derivative, whose components have lower degree. Proposition 7.1 transfers this estimate back to the original law, with an error involving only lower-degree coefficients.

The covariance bound starts the induction at degree one. Letwin’s degree-two bound and the operator lemma control one integration for the small-degree steps. For large degrees, we integrate about half the derivatives at once. The decay built into the coefficient bound lets the power lemma supply a strict gain at a fixed exponential rate. Section 8 states the resulting bound, and Appendix E verifies the induction with explicit constants.

Finally, fix a regular measure and apply the operator estimate with $q = 1$. Letting $D \rightarrow \infty$ removes the curvature factor. The decomposition into constant and centered inputs then gives the Poincaré inequality. Section 9 passes it to general log-concave measures and deduces the Cheeger bound.

The below diagram summarizes the induction and the final deduction. The covariance bound starts the induction at degree one. Bounds in lower degrees enter both the power estimate and the error in reverse transfer.



For a worked example illustrating the proof ideas for a planar radial family, see:

<https://kriznakumar.github.io/blog/>

1.2 AI Usage Statement

We developed this proof with substantial assistance from several frontier AI models. The project was inspired by the elegance of Letwin’s approach (Letwin, 2026). Our earlier argument based on a variational Stein-kernel formulation with a structured tensor recovery from partial symmetries bound, yielded a bound for the reciprocal Cheeger constant with a fourth-root dependence on the iterated logarithm of the dimension. The present proof retains Letwin’s quadratic variance estimate and uses compatible integration operators to close a direct induction on polynomial degree. Two AI interactions are worth emphasizing:

- Motivated by the use of Helmholtz–Hodge decomposition in the analysis of drifting models (Balasubramanian, 2026), we asked the AI whether a similar decomposition could help control curl in our proposed KLS argument. We suggested decomposing the weighted divergence of a compatible symmetric tensor field into its compatible component and an orthogonal remainder, then bounding the squared L^2 norm of that remainder. The AI identified the need for estimates uniform in tensor rank and formulated a weighted Hodge comparison for symmetric tensor fields.
- We also asked the AI whether the Fourier coefficient strategy of Balasubramanian et al. (2025) could inform the KLS argument. We suggested organizing the functional-analytic problem around a tractable hierarchy of coefficients controlled by recursive bounds. The AI proposed Appell coefficient norms to measure repeated centered integration of constant tensors. It also proposed using curvature and the Hodge estimate to transfer these bounds to repeated integration of arbitrary compatible fields with mean zero.

We have carefully verified all arguments developed with the assistance of AI and take full responsibility for the content of this work. We also believe that the argument may admit further simplification, possibly through other techniques or intuitions already familiar to the community.

2 Preliminaries

We fix notation and regularity assumptions, define the Appell coefficient norms, and state Letwin’s quadratic variance estimate.

2.1 Notation and Regularity

The operator arguments require weighted function spaces and a smooth class of measures. We fix these conventions here, along with the tensor norms used throughout the proof.

For a finite-dimensional Euclidean space E , $L^p(\mu; E)$ is the space of measurable E -valued functions with finite p th norm ($1 \leq p < \infty$). The norm in $L^\infty(\mu; E)$ uses the essential supremum. For an integer $k \geq 0$, the weighted Sobolev space $W^{k,2}(\mu; E)$ consists of functions whose weak derivatives through order k are square-integrable under μ . We use the Euclidean norm on their components. We omit E when it is $\mathbb{R}$ or clear from the field being discussed. The subscript zero in $L_0^2(\mu)$ means that the function has mean zero. The spaces $C_c^\infty(\mathbb{R}^n; E)$ consist of smooth compactly supported E -valued functions.

All tensor spaces carry the Euclidean Hilbert–Schmidt norm: the squared norm is the sum of squared entries over ordered indices. For $r \in \mathbb{N}_0 = \{0, 1, \dots\}$, write $\text{Sym}^r \mathbb{R}^n \subseteq (\mathbb{R}^n)^{\otimes r}$ for the symmetric tensors of rank r , with $(\mathbb{R}^n)^{\otimes 0} = \mathbb{R}$. We identify $\text{Sym}^2 \mathbb{R}^n$ with the real symmetric

$n \times n$ matrices. For $T \in (\mathbb{R}^n)^{\otimes r}$, $\text{Sym } T$ is the average over all permutations of its indices. Thus, $\text{Sym} : (\mathbb{R}^n)^{\otimes r} \rightarrow \text{Sym}^r \mathbb{R}^n$ is the orthogonal projection. The vectors $e_i \in \mathbb{R}^n$, $1 \leq i \leq n$, denote the standard basis. We use $\preceq$ for the order on self-adjoint operators, defined by comparison of their quadratic forms. For a Hilbert-space operator T , T^* is its adjoint and $\text{Dom}(T)$ is its domain. If $\mathcal{M}$ is a closed subspace of a Hilbert space $\mathcal{X}$, then $P_{\mathcal{M}} : \mathcal{X} \rightarrow \mathcal{M}$ denotes orthogonal projection. The symbol Id denotes the identity on the space in question. All logarithms are natural. Expectations without a subscript refer to the measure currently under consideration.

Let $B \in \text{Sym}^2 \mathbb{R}^n$ be positive semidefinite. A measure has *curvature at least B* if its potential can be written as $V(x) = W(x) + x^\top Bx/2$. Here, W is extended-real-valued and convex on the affine support. This definition applies to measures with convex support as well as to smooth densities. In the smooth case it is equivalent to $D^2V \succeq B$.

Definition 2.1 (Regular Law). A centered probability measure on $\mathbb{R}^n$ is *regular* if it has density $Z^{-1}e^{-V(x)}$, with $V \in C^\infty(\mathbb{R}^n)$ and

$$0 < a\text{Id} \preceq D^2V(x) \preceq a_+\text{Id}$$

for some $a, a_+ \in (0, \infty)$. Here, $Z = \int_{\mathbb{R}^n} e^{-V(x)} dx \in (0, \infty)$ normalizes the density. Both Hessian bounds may depend on the measure.

The integration operators are first defined for regular measures with covariance at most Id . Later approximation removes the regularity assumption. The ordinary Bakry–Émery inequality (Bakry et al., 2014, Chapter 4) gives

$$C_P(\mu) \leq a^{-1} \quad \text{when } D^2V \succeq a\text{Id}, \quad a > 0. \quad (2.1)$$

2.2 Appell Coefficients

Appell polynomials are adapted to integration with mean zero. We define their normalization and coefficient norms, and record how the polynomials behave under differentiation, affine changes of variables, and limits of moments. These facts will connect integration bounds to the induction on polynomial degree.

Let μ be a probability measure on $\mathbb{R}^n$ with all moments, and let X be an $\mathbb{R}^n$ -valued random vector with law μ . We write $\mathbb{R}[x_1, \dots, x_n]$ for the space of real polynomials on $\mathbb{R}^n$. The Appell tensors of μ are polynomial maps $\mathcal{A}_d^\mu : \mathbb{R}^n \rightarrow \text{Sym}^d \mathbb{R}^n$, $d \in \mathbb{N}_0$. They are defined by the formal identity (Anshelevich, 2004, Section 2.3, Equation (11))

$$\frac{e^{z \cdot x}}{\mathbb{E}_\mu e^{z \cdot X}} = \sum_{d=0}^{\infty} \frac{1}{d!} \langle \mathcal{A}_d^\mu(x), z^{\otimes d} \rangle. \quad (2.2)$$

Here, $x \in \mathbb{R}^n$, and $z_1, \dots, z_n$ are formal variables. The identity is in $\mathbb{R}[[z_1, \dots, z_n]]$, the ring of formal power series. Each coefficient uses only finitely many moments, so the definition does not require convergence of the series. For $T \in \text{Sym}^d \mathbb{R}^n$, the scalar polynomial $P_d^\mu[T] : \mathbb{R}^n \rightarrow \mathbb{R}$ and its normalized coefficient bound $c_d(\mu) \in [0, \infty)$ are

$$P_d^\mu[T] = \langle T, \mathcal{A}_d^\mu \rangle, \quad c_d(\mu) = \frac{1}{d!} \sup_{\substack{T \in \text{Sym}^d \mathbb{R}^n \\ \|T\|=1}} \|P_d^\mu[T]\|_{L^2(\mu)}. \quad (2.3)$$

For $d \geq 1$, we write

$$c_d^* = \sup_{\mu} c_d(\mu) \in [0, \infty], \quad (2.4)$$

taking the supremum over all dimensions and all centered log-concave laws with covariance at most Id. These uniform coefficients are proved finite in Theorem 8.1. For $d \geq 1$, taking expectations in Equation (2.2) gives $\mathbb{E}_{\mu} \mathcal{A}_d^{\mu} = 0$. For $v \in \mathbb{R}^n$, let $D_v = v \cdot \nabla$ be the directional derivative on scalar polynomials. Differentiation gives

$$D_v P_d^{\mu}[T] = d P_{d-1}^{\mu}[T(v, \cdot, \dots, \cdot)]. \quad (2.5)$$

Thus, every expected derivative of order less than d vanishes. This is the property that makes the polynomials compatible with repeated centered integration.

For example, if μ is centered and has covariance $\Sigma \in \text{Sym}^2 \mathbb{R}^n$, then

$$\mathcal{A}_1^{\mu}(x) = x, \quad \mathcal{A}_2^{\mu}(x) = x^{\otimes 2} - \Sigma,$$

and

$$\mathcal{A}_3^{\mu}(x) = x^{\otimes 3} - 3 \text{Sym}(x \otimes \Sigma) - \mathbb{E}_{\mu} X^{\otimes 3}.$$

The covariance correction in the cubic tensor makes its expected gradient zero, while the final constant centers its values.

For changes of coordinates, we use the following affine transformation rule. If X and Y take values in $\mathbb{R}^n$ and $\mathbb{R}^{\ell}$, respectively, and $X = m + BY$ with $m \in \mathbb{R}^n$ and $B \in \mathbb{R}^{n \times \ell}$, coefficient comparison in Equation (2.2) gives

$$P_d^{\mathcal{L}(X)}[T](m + By) = P_d^{\mathcal{L}(Y)}[(B^{\top})^{\otimes d} T](y). \quad (2.6)$$

Here, $\mathcal{L}(X)$ denotes the probability law of X , and $(B^{\top})^{\otimes d}$ applies $B^{\top}$ in every tensor slot.

For approximation, we use continuity in the moments of the law. The coefficients of $\mathcal{A}_d^{\mu}$ are polynomials in the moments of μ through order d , obtained by inverting the formal series in Equation (2.2). Their squared L^2 norms and derivative norms therefore depend continuously on moments through order $2d$. The derivative $D^q P_d^{\mu}[T]$ is a polynomial map $\mathbb{R}^n \rightarrow \text{Sym}^q \mathbb{R}^n$. In a fixed dimension, if those moments converge from probability measures μ_j to μ , then

$$\begin{aligned} c_d(\mu_j) &\longrightarrow c_d(\mu), \\ \|D^q P_d^{\mu_j}[T]\|_{L^2(\mu_j)}^2 &\longrightarrow \|D^q P_d^{\mu}[T]\|_{L^2(\mu)}^2 \quad (0 \leq q \leq d). \end{aligned} \quad (2.7)$$

For the first limit, use entrywise convergence of the Gram matrix of the degree- d Appell tensors in an orthonormal basis of $\text{Sym}^d \mathbb{R}^n$. Its largest eigenvalue is $(d!c_d(\mu))^2$. In a fixed dimension the matrix has fixed finite size, so entrywise convergence implies operator-norm convergence and convergence of this eigenvalue. Thus, the supremum over leading tensors also passes to the limit.

2.3 The Quadratic Variance Input

We control covariance fluctuations during localization using the quadratic variance inequality of Letwin (2026, Theorem 1.2, Equation (1.2)). Every isotropic log-concave random vector X in $\mathbb{R}^n$ satisfies

$$\text{Var}(X^T B X) \leq 8 \|B\|_{\text{HS}}^2 \quad \text{for every } B \in \text{Sym}^2 \mathbb{R}^n. \quad (2.8)$$

Letwin's formulation is $\text{Var}(f(X)) \leq 2\mathbb{E}|\nabla f(X)|^2$ for quadratic forms $f(x) = x^T Bx$. The displayed normalization follows from $\nabla f(x) = 2Bx$ and $\mathbb{E}|BX|^2 = \|B\|_{\text{HS}}^2$. Since $P_2^\mu[B] = X^T B X - \mathbb{E}[X^T B X]$, dividing by $2!$ gives

$$c_2^* \leq \sqrt{2}. \quad (2.9)$$

For a centered law with covariance at most Id , the same estimate follows by writing it as a linear contraction of an isotropic law on its support and using Equation (2.6). A point mass has zero positive-degree Appell coefficients. In Section 6, we apply this inequality to the isotropic coordinates of each localized law and derive the third-moment estimates in Equation (6.4).

We also use two basic facts in localization and approximation. All log-concave laws have finite moments of each fixed order. The tail argument at the start of Appendix F proves this directly. Prékopa's marginal theorem (Prékopa, 1973, Theorem 6) shows that integrating out variables preserves log-concavity. We use it when smoothing a strongly log-concave measure by Gaussian convolution. We also emphasize that $c_d(\mu)$ belongs to one law, c_d^* is a supremum over laws, and C_k denotes a chosen upper bound in reverse transfer. The indices r and d refer to tensor rank and polynomial degree, respectively.

3 A Hodge Estimate for Compatible Symmetric Tensors

In this section, we state the curvature estimate used to control integration and illustrate it in the smallest matrix case. The proof is given in Appendix A.

Throughout this section, μ is regular in the sense of Definition 2.1, and $K = D^2V : \mathbb{R}^n \rightarrow \text{Sym}^2 \mathbb{R}^n$ is its Hessian field. The upper bound a_+ is used in the domain and approximation arguments; the final estimate depends only on the lower bound a . Tensor norms sum squares over all ordered tensor indices; exterior-form norms sum squares over strictly increasing indices. On $C_c^\infty(\mathbb{R}^n)$, write

$$\partial_i^* = -\partial_i + \partial_i V, \quad H = \sum_i \partial_i^* \partial_i.$$

Here, ∂_i^* is the weighted adjoint of the coordinate derivative in $L^2(\mu)$. We use H for the nonnegative self-adjoint operator on $L^2(\mu)$ associated with the energy $f \mapsto \int |\nabla f|^2 d\mu$, whose form domain is $W^{1,2}(\mu)$. Its operator domain consists of the $f \in W^{1,2}(\mu)$ for which $Hf \in L^2(\mu)$ in distributions. It acts componentwise on tensor fields. Integration by parts gives $[\partial_i, \partial_j^*] = V_{ij}$, where $V_{ij} = \partial_i \partial_j V$.

3.1 Compatible Fields and the Estimate

To formulate the Hodge estimate, we need tensor fields that can be integrated repeatedly. We define those fields and the derivative acting on them, then state the estimate and work through the smallest matrix example.

Let $E_r = \text{Sym}^r(\mathbb{R}^n)$, with $E_0 = \mathbb{R}$. A symmetric rank- r field $F \in L^2(\mu; E_r)$, $r \geq 1$, is called compatible if

$$\partial_i F_{jI} = \partial_j F_{iI}. \quad (3.1)$$

The identity holds in distributions for every ordered index tuple $I \in \{1, \dots, n\}^{r-1}$. These are the relations satisfied by repeated derivatives of a scalar function. Let $\mathcal{C}_r \subset L^2(\mu; E_r)$ be the closed space of compatible fields, and put

$$\mathcal{G}_r = \{F \in \mathcal{C}_r : \mathbb{E}F = 0\}.$$

For rank zero, set $\mathcal{C}_0 = L^2(\mu)$ and $\mathcal{G}_0 = L^2_0(\mu)$. Constant tensors are compatible, so

$$\mathcal{C}_r = E_r \oplus \mathcal{G}_r$$

orthogonally. We call the gradient *raw* when its output is not centered by subtracting its mean. It is the closed, densely defined operator

$$\mathcal{D}_r : \mathcal{G}_r \longrightarrow \mathcal{C}_{r+1}, \quad \text{Dom}(\mathcal{D}_r) = \mathcal{G}_r \cap W^{1,2}(\mu; E_r), \quad \mathcal{D}_r U = \nabla U. \quad (3.2)$$

Compatibility ensures that ∇U is fully symmetric and belongs to the stated target space. Its Hilbert-space adjoint has domain in $\mathcal{C}_{r+1}$ and values in $\mathcal{G}_r$: $\mathcal{D}_r^* : \text{Dom}(\mathcal{D}_r^*) \subseteq \mathcal{C}_{r+1} \rightarrow \mathcal{G}_r$.

For the smooth fields $F = \nabla^{r+1} \psi \in C_c^\infty(\mathbb{R}^n; E_{r+1})$, $\psi \in C_c^\infty(\mathbb{R}^n)$, the weighted divergence $\text{div}_V F \in C_c^\infty(\mathbb{R}^n; E_r)$ is defined by

$$(\text{div}_V F)_I = \sum_j \partial_j^* F_{Ij}. \quad (3.3)$$

Weighted integration by parts gives $\mathbb{E} \text{div}_V F = 0$ and

$$\mathcal{D}_r^* F = P_{\mathcal{G}_r} \text{div}_V F = P_{\mathcal{C}_r} \text{div}_V F. \quad (3.4)$$

The two projections agree here because constant tensors belong to $\mathcal{C}_r$ and are orthogonal to $\text{div}_V F$.

The lemma below controls both a compatible tensor field and its first derivative. The curvature term has the same lower bound a at every rank, so adding tensor indices does not weaken the estimate.

Lemma 3.1 (Compatible-Tensor Hodge Estimate). *For every integer $m \geq 0$ and every $F \in \text{Dom}(\mathcal{D}_m^*)$, one has $F \in W^{1,2}(\mu; E_{m+1})$ and*

$$\|\mathcal{D}_m^* F\|_2^2 \geq \|\nabla F\|_2^2 + a \|F\|_2^2. \quad (3.5)$$

The Case $n = 2$, $m = 1$. Take $F = D^2 \psi$ with $\psi \in C_c^\infty(\mathbb{R}^2)$, so F is a compatible symmetric matrix field in $C_c^\infty(\mathbb{R}^2; \text{Sym}^2 \mathbb{R}^2)$. Put $Y = \text{div}_V F \in C_c^\infty(\mathbb{R}^2; \mathbb{R}^2)$. At a fixed point, choose a basis in which $K = \text{diag}(\kappa_1, \kappa_2)$. Equation (A.2) gives the scalar curl $t_{12} \in C_c^\infty(\mathbb{R}^2)$:

$$t_{12} = \partial_1 Y_2 - \partial_2 Y_1 = (\kappa_1 - \kappa_2) F_{12}.$$

The two-form space is one-dimensional, and its curvature operator is multiplication by $\kappa_1 + \kappa_2$. The projection estimate in Appendix A therefore gives

$$\|(\text{Id} - P_{\mathcal{C}_1})Y\|_2^2 \leq \mathbb{E} \frac{(\kappa_1 - \kappa_2)^2 F_{12}^2}{\kappa_1 + \kappa_2}.$$

The mixed component contributes $(\kappa_1 + \kappa_2) F_{12}^2$ to the original curvature form. Subtracting the displayed loss leaves

$$\left(\kappa_1 + \kappa_2 - \frac{(\kappa_1 - \kappa_2)^2}{\kappa_1 + \kappa_2} \right) F_{12}^2 = \frac{2\kappa_1 \kappa_2}{\kappa_1 + \kappa_2} (2F_{12}^2).$$

Thus, the remaining weight is the harmonic mean, multiplied by the squared norm $2F_{12}^2$ of the mixed symmetric component. The loss bound is zero when $\kappa_1 = \kappa_2$; unequal curvatures account for the projection error.

4 Compatible Integration

In this section, we construct the integration operators used in the proof. At each integration, we choose the additive constant to give mean zero. Starting from a constant tensor produces the normalized Appell tensors. Curvature estimates control integration on all compatible fields, whereas Appell coefficient bounds describe what happens when the starting field is constant. We put these two estimates on the same Hilbert space. In the next section, we use bounds through a finite polynomial degree to control any number of integrations.

Throughout this section, ν is regular as in Definition 2.1, with $\text{Cov}(\nu) \preceq \text{Id}$ and $D^2V \succeq a\text{Id}$ for some $a > 0$. All norms and expectations refer to ν unless otherwise indicated.

4.1 Centered Primitives

Choosing mean zero fixes the additive constant in a primitive. We use this choice to define integration operators and relate their norms to the Poincaré constant.

Lemma B.1 says that the derivative in Equation (3.2) is bijective and has a bounded inverse

$$\tilde{J}_r = \mathcal{D}_r^{-1} : \mathcal{G}_{r+1} \longrightarrow \mathcal{G}_r.$$

Thus, $\nabla \tilde{J}_r F = F$ and $\mathbb{E} \tilde{J}_r F = 0$. The construction uses the distributional Poincaré lemma to obtain a primitive and the scalar Poincaré inequality to place its centered representative in $L^2(\nu)$. In particular, $\|\tilde{J}_r\| \leq \sqrt{C_P(\nu)}$.

Separate the constant and centered parts of the input by defining

$$\begin{aligned} J_r &= \tilde{J}_r|_{\mathcal{G}_{r+1}} : \mathcal{G}_{r+1} \longrightarrow \mathcal{G}_r, \\ L_r : E_{r+1} &\longrightarrow \mathcal{G}_r, \quad (L_r T)(x) = T \lrcorner x. \end{aligned}$$

Here, $T \in E_{r+1}$ and $x \in \mathbb{R}^n$; contraction against x leaves r free indices. Since $\mathbb{E}X = 0$, $L_r T$ is the centered primitive of the constant field T . The positive self-adjoint operator associated with the gradient energy is

$$H_r = \mathcal{D}_r^* \mathcal{D}_r : \text{Dom}(H_r) \subset \mathcal{G}_r \longrightarrow \mathcal{G}_r,$$

where $\text{Dom}(H_r) = \{U \in \text{Dom}(\mathcal{D}_r) : \mathcal{D}_r U \in \text{Dom}(\mathcal{D}_r^*)\}$. The inverse of H_r is $\tilde{J}_r \tilde{J}_r^*$: indeed, $(\mathcal{D}_r^*)^{-1} = (\mathcal{D}_r^{-1})^*$, and the inverses multiply in the reverse order. Splitting the input space of $\tilde{J}_r$ into its two orthogonal summands therefore gives

$$H_r^{-1} = \tilde{J}_r \tilde{J}_r^* = J_r J_r^* + L_r L_r^*. \quad (4.1)$$

The covariance assumption controls the constant part uniformly in rank:

$$\|L_r T\|_2^2 = \sum_{i_1, \dots, i_r} \mathbb{E} \left(\sum_j T_{i_1 \dots i_r j} X_j \right)^2 \leq \sum_{i_1, \dots, i_r, j} T_{i_1 \dots i_r j}^2 = \|T\|^2. \quad (4.2)$$

The scalar Poincaré inequality controls all ranks. If $F \in \mathcal{G}_{r+1}$ and $U = J_r F \in \mathcal{G}_r \cap W^{1,2}(\nu; E_r)$, then every component U_I satisfies $\mathbb{E}U_I = 0$. Applying that inequality to each component and using $\nabla U = F$ gives

$$\|J_r F\|_2^2 \leq C_P(\nu) \|F\|_2^2, \quad \|J_r\|_{\text{op}}^2 \leq C_P(\nu) \quad (r \geq 0). \quad (4.3)$$

4.2 Compatibility Projection and Curvature

Lemma 3.1 gives, for every $F \in \text{Dom}(\mathcal{D}_r^*)$,

$$\|\mathcal{D}_r^* F\|_2^2 \geq \|\nabla F\|_2^2 + a \|F\|_2^2.$$

Its assertion about the adjoint domain also gives $F \in W^{1,2}(\nu; E_{r+1})$. Taking inverses will turn this derivative bound into a comparison of integration operators. This is the curvature input to Lemma 5.1.

Let $\tilde{H}_{r+1} : \text{Dom}(\tilde{H}_{r+1}) \subset \mathcal{C}_{r+1} \rightarrow \mathcal{C}_{r+1}$ be the self-adjoint operator associated with the gradient form on the full space $\mathcal{C}_{r+1}$. Relative to $\mathcal{C}_{r+1} = E_{r+1} \oplus \mathcal{G}_{r+1}$, it is the direct sum of zero and H_{r+1} , with domain $E_{r+1} \oplus \text{Dom}(H_{r+1})$. The Hodge estimate, including its assertion about the form domains, reads

$$\mathcal{D}_r \mathcal{D}_r^* \succeq \tilde{H}_{r+1} + a \text{Id}. \quad (4.4)$$

Inversion reverses the order of positive operators. Let T be a self-adjoint operator on $\mathcal{C}_{r+1}$ with a positive lower bound. For $f \in \mathcal{C}_{r+1}$, the variational formula is

$$\langle f, T^{-1} f \rangle = \sup_u \{ 2 \text{Re} \langle f, u \rangle - \|T^{1/2} u\|^2 \},$$

where the supremum is over $u \in \text{Dom}(T^{1/2})$, the form domain. A larger form, on a possibly smaller domain, gives a smaller supremum. Applying this to Equation (4.4) yields

$$\tilde{J}_r^* \tilde{J}_r \preceq (\tilde{H}_{r+1} + a \text{Id})^{-1}.$$

Restricting this inequality to inputs in $\mathcal{G}_{r+1}$ gives $J_r^* J_r \preceq (H_{r+1} + a \text{Id})^{-1}$. Functional calculus and Equation (4.1) now give the operator inequality that will be used below:

$$J_r^* J_r \preceq g(J_{r+1} J_{r+1}^* + L_{r+1} L_{r+1}^*), \quad g(u) = \frac{u}{1 + au}. \quad (4.5)$$

Indeed, $(H_{r+1} + a \text{Id})^{-1} = g(H_{r+1}^{-1})$. Here, $g : [0, \infty) \rightarrow [0, a^{-1})$ is applied to operators by spectral calculus: it acts on each spectral value as the displayed scalar function. The argument of g in Equation (4.5) is a bounded positive operator of norm at most $C_P(\nu) + 1$, by Equations (4.2) and (4.3).

4.3 Appell Tensors from Repeated Integration

Starting from a constant tensor, repeated centered integration gives the Appell normalization in Equation (2.2). We verify this identity and then use the coefficient norms to bound the resulting products of integration operators. For every $r, k \geq 0$ and $T \in E_{r+k+1}$,

$$J_r J_{r+1} \cdots J_{r+k-1} L_{r+k} T = \frac{1}{(k+1)!} T \lrcorner \mathcal{A}_{k+1}^\nu. \quad (4.6)$$

Both sides belong to $\mathcal{G}_r$, and the contraction on the right has r free indices. For $k = 0$, the left side means $L_r T$, and the equality is $\mathcal{A}_1^\nu(x) = x$. For the induction step, differentiating the right side gives

$$\nabla \left(\frac{1}{(k+1)!} T \lrcorner \mathcal{A}_{k+1}^\nu \right) = \frac{1}{k!} T \lrcorner \mathcal{A}_k^\nu,$$

now with $r + 1$ free indices. When $k \geq 1$, this derivative has mean zero because $\mathbb{E}\mathcal{A}_k^\nu = 0$. The original field also has mean zero, and its full derivative is symmetric, so it is compatible. It is therefore the unique centered primitive selected by J_r . The induction hypothesis applied at rank $r + 1$ proves Equation (4.6).

To bound the norm in this identity, fix an ordered free-index tuple I . Let $T_I \in E_{k+1}$ denote the remaining symmetric rank- $(k + 1)$ slice of T . The definition in Equation (2.3) gives

$$\left\| \frac{\langle T_I, \mathcal{A}_{k+1}^\nu \rangle}{(k+1)!} \right\|_2^2 \leq c_{k+1}(\nu)^2 \|T_I\|^2.$$

Summing over I and using $\sum_I \|T_I\|^2 = \|T\|^2$ proves

$$\|J_r \cdots J_{r+k-1} L_{r+k}\|_{\text{op}} \leq c_{k+1}(\nu). \quad (4.7)$$

4.4 Integration Across Tensor Ranks

So far, we have considered integration at each rank separately. Putting them on one Hilbert space lets us express the curvature comparison and coefficient bounds as the two hypotheses of the operator lemma. For this purpose, introduce the Hilbert direct sums

$$\mathcal{H} = \bigoplus_{r \geq 0} \mathcal{G}_r, \quad \mathcal{E} = \bigoplus_{r \geq 0} E_{r+1},$$

whose elements have square-summable component norms. Define the bounded operators $J : \mathcal{H} \rightarrow \mathcal{H}$ and $L : \mathcal{E} \rightarrow \mathcal{H}$ by

$$(JF)_r = J_r F_{r+1}, \quad (LU)_r = L_r U_r. \quad (4.8)$$

Here, $F = (F_r)_{r \geq 0} \in \mathcal{H}$ and $U = (U_r)_{r \geq 0} \in \mathcal{E}$. Thus, J lowers the rank by one, while L sends constant tensors to their centered linear primitives. The bounds at each rank give

$$\|J\|_{\text{op}}^2 \leq C_P(\nu) < \infty, \quad \|L\|_{\text{op}} \leq 1. \quad (4.9)$$

The norm of J is the supremum of $\|J_r\|$, and each block is bounded by $\sqrt{C_P(\nu)}$.

Conversely, at rank zero the energy of H_0 is the Poincaré variational form on $L_0^2(\nu)$, whose least spectral value is $C_P(\nu)^{-1}$. Equations (4.1) and (4.2), together with the fact that J_0 is a block of J , give

$$\begin{aligned} C_P(\nu) &= \|H_0^{-1}\|_{\text{op}} \\ &\leq \|J_0\|_{\text{op}}^2 + \|L_0\|_{\text{op}}^2 \leq 1 + \|J\|_{\text{op}}^2. \end{aligned} \quad (4.10)$$

Thus, a bound for integration on centered inputs controls the full Poincaré constant.

Both JJ^* and LL^* are block-diagonal, with blocks $J_r J_r^*$ and $L_r L_r^*$ on $\mathcal{G}_r$. Functional calculus therefore acts on each summand separately. For $r \geq 1$, Equation (4.5) at index $r - 1$ gives

$$\begin{aligned} (J^* J)|_{\mathcal{G}_r} &= J_{r-1}^* J_{r-1} \\ &\preceq g(J_r J_r^* + L_r L_r^*) \\ &= g(JJ^* + LL^*)|_{\mathcal{G}_r}. \end{aligned}$$

On $\mathcal{G}_0$, the left side is zero and the right side is positive. Combining the blocks gives

$$J^* J \preceq g(JJ^* + LL^*). \quad (4.11)$$

For $j \geq 1$, the nonzero blocks of $J^{j-1}\mathbf{L}$ are

$$J_r J_{r+1} \cdots J_{r+j-2} L_{r+j-1}, \quad r \geq 0,$$

with the expression interpreted as L_r when $j = 1$. Their source and target summands are mutually orthogonal. Therefore, Equation (4.7) implies

$$\left\| J^{j-1}\mathbf{L} \right\|_{\text{op}} \leq c_j(\nu). \quad (4.12)$$

Equations (4.11) and (4.12) are the inputs to the abstract argument in the next section.

5 An Operator Lemma

In this section, we consider bounded operators $R : \mathcal{X} \rightarrow \mathcal{X}$ and $L : \mathcal{Y} \rightarrow \mathcal{X}$ between Hilbert spaces. The following lemma shows how bounds on finitely many products $R^{j-1}L$ can control every power of R . The same multiplicative constant works for all powers. We apply this result to repeated integration in Section 5.1.

Lemma 5.1 (Uniform Power Bound). *Let $R : \mathcal{X} \rightarrow \mathcal{X}$ and $L : \mathcal{Y} \rightarrow \mathcal{X}$ be bounded operators on Hilbert spaces, and let $a > 0$. Suppose*

$$R^*R \preceq g(RR^* + LL^*), \quad g(u) = \frac{u}{1 + au}. \quad (5.1)$$

Let $D \geq 2$ be an integer and $\gamma_2, \dots, \gamma_D$ be finite nonnegative numbers such that

$$\left\| R^{j-1}L \right\|_{\text{op}} \leq \gamma_j \quad (2 \leq j \leq D).$$

For every $B > 0$ with $aB^{D+1} > \gamma_D^2$ and every integer $k \geq 0$,

$$\left\| R^k \right\|_{\text{op}}^2 \leq B^k Q_D(B), \quad Q_D(B) = 1 + \sum_{j=1}^{D-2} \frac{j\gamma_{j+1}^2}{B^{j+1}}. \quad (5.2)$$

The sum is empty when $D = 2$. Here, $g : [0, \infty) \rightarrow [0, a^{-1})$ is the scalar function applied by spectral calculus, and $Q_D(B) \in [1, \infty)$ is the common prefactor.

The contribution of L is essential to the hypothesis. For example, if $L = 0$, taking norms in Equation (5.1) gives $\|R\|^2 \leq g(\|R\|^2)$, and hence $R = 0$ because $g(u) < u$ for $u > 0$. The observations measure how this contribution behaves under iteration.

Proof Idea. The operator inequality and the last observation, $\|R^{D-1}L\| \leq \gamma_D$, first bound the spectral radius of R strictly below $\sqrt{B}$. For a unit vector $f \in \mathcal{X}$, the scaled adjoint orbit $B^{-m}\|(R^*)^m f\|^2$ therefore tends to zero and attains a maximum. The same inputs yield a recurrence that forces this maximum to occur before index $D - 1$. The remaining observation bounds control its size by $Q_D(B)$, so the same prefactor works for every power. Appendix C gives the spectral argument and the recurrence in full.

5.1 Application to the Graded Integration Operator

Let ν be a regular law with $\text{Cov}(\nu) \preceq \text{Id}$ and $D^2V \succeq a\text{Id}$, where $a > 0$. Both corollaries below apply Lemma 5.1 with $R = J$, $L = L$, and finite observation bounds $\gamma_j \geq c_j(\nu)$ for $2 \leq j \leq D$, where $D \geq 2$ is an integer. Equations (4.11) and (4.12) verify its operator and observation hypotheses. The first corollary uses coefficient bounds through degree D to control any number of integrations with prefactor 2. The second uses only the quadratic estimate to supply the initial integration bound.

Corollary 5.2 (Uniform Integration Powers). *Let ν be regular with $\text{Cov}(\nu) \preceq \text{Id}$ and curvature at least $a\text{Id}$ for some $a > 0$. Let $D \geq 2$ be an integer and $\rho \geq 1$. Suppose*

$$c_j(\nu) \leq \frac{\rho^j}{(j+1)^4} \quad (2 \leq j \leq D).$$

Put

$$B = \rho^2 \max\{1, a^{-1/(D+1)}\}.$$

Then, simultaneously for every integer $q \geq 1$,

$$\|J^q\|_{\text{op}}^2 \leq 2B^q, \quad \|J_0 J_1 \cdots J_{q-1}\|_{\text{op}}^2 \leq 2B^q. \quad (5.3)$$

In particular, $C_P(\nu) \leq 1 + 2B$.

Proof. The choice of B , together with $\rho \geq 1$ and $D \geq 2$, gives

$$aB^{D+1} \geq \rho^{2D+2} > \frac{\rho^{2D}}{(D+1)^8} \geq c_D(\nu)^2.$$

Also $B \geq \rho^2$, so

$$\begin{aligned} Q_D(B) - 1 &\leq \sum_{j=1}^{D-2} \frac{j}{(j+2)^8} \leq \sum_{j=1}^{\infty} \frac{1}{(j+2)^7} \\ &\leq \int_2^{\infty} x^{-7} dx = \frac{1}{384} < 1. \end{aligned}$$

Lemma 5.1 gives the first bound in Equation (5.3). The product $J_0 \cdots J_{q-1}$ is the block of J^q from $\mathcal{G}_q$ to $\mathcal{G}_0$, so its norm is no larger. Finally take $q = 1$ and use Equation (4.10). $\square$

Corollary 5.3 (Quadratic Seed). *Let ν be regular with $\text{Cov}(\nu) \preceq \text{Id}$ and curvature at least $a\text{Id}$, where $a > 0$. Then*

$$\|J\|_{\text{op}}^2 \leq (2/a)^{1/3}. \quad (5.4)$$

Proof. Equation (2.9) gives $c_2(\nu) \leq \sqrt{2}$. Choose $D = 2$ and $\gamma_2 = \sqrt{2}$. The prefactor sum is empty, so $Q_2(B) = 1$. For every $B > (2/a)^{1/3}$, Lemma 5.1 gives $\|J\|^2 \leq B$. Let B decrease to $(2/a)^{1/3}$. $\square$

The seed uses only Letwin's quadratic estimate and the operator lemma; it does not assume any higher-degree coefficient bound.

5.2 The Gaussian Equality Case

For a one-dimensional law, the centered integration operators at all ranks identify with a scalar operator J on the mean-zero functions. Write $L : \mathbb{R} \rightarrow L_0^2(\nu)$ for the constant-input primitive, so $L(u)(x) = u(x - \mathbb{E}_\nu X)$. The standard Gaussian gives equality in the curvature comparison.

Let $\nu = N(0, 1)$ be the standard Gaussian law on $\mathbb{R}$ and $h_j = \text{He}_j / \sqrt{j!} \in L^2(\nu)$ its normalized Hermite polynomials, where $\text{He}_j : \mathbb{R} \rightarrow \mathbb{R}$ is the degree- j probabilists' Hermite polynomial (Anshelevich, 2004, Section 2.7). The functions h_j , $j \geq 1$, form an orthonormal basis of $L_0^2(\nu)$. Since $h'_{j+1} = \sqrt{j+1} h_j$,

$$Jh_j = \frac{h_{j+1}}{\sqrt{j+1}}, \quad L(1) = h_1.$$

Thus, on this basis,

$$J^* J h_j = \frac{h_j}{j+1}, \quad (JJ^* + LL^*)h_j = \frac{h_j}{j} \quad (j \geq 1).$$

For $a = 1$, the function $g(u) = u/(1+u)$ sends $1/j$ to $1/(j+1)$. Therefore, the comparison in Equation (4.5) is an equality at every rank. In the same normalization, $c_d(\nu) = 1/\sqrt{d!}$.

6 Stochastic Localization

To use the integration bounds, we localize the measure to create positive curvature. We need two estimates along this process: one for covariance and one for the moving Appell variance. The covariance estimate controls the cost of the final change of coordinates. The variance estimate compares the original polynomial with the polynomial of the localized measure. Section 7 combines these two estimates with the integration bound.

6.1 Localization and Mixed Covariance Moments

We first work with an isotropic compactly supported log-concave law μ_0 on $\mathbb{R}^n$. For this law, we construct localization and bound products of covariance matrices, including matrices evaluated at different times. These estimates will control the changes of coordinates in reverse transfer.

Choose $R > 0$ such that the support is contained in the closed ball $B_R = \{x \in \mathbb{R}^n : |x| \leq R\}$. We use the covariance-normalized instance of stochastic localization from Eldan (2013, Section 2, equations (11)–(13)), within the matrix-driven formulation of Chen and Eldan (2025, Section 2.4.2, Equation (5) and Fact 14).

For $\theta \in \mathbb{R}^n$ and $\Lambda \in \text{Sym}^2 \mathbb{R}^n$, define the probability law

$$\mu_{\theta, \Lambda}(dx) = \frac{e^{\theta \cdot x - x^T \Lambda x / 2}}{\int e^{\theta \cdot y - y^T \Lambda y / 2} \mu_0(dy)} \mu_0(dx).$$

Its mean $m(\theta, \Lambda) \in \mathbb{R}^n$ and covariance $A(\theta, \Lambda) \in \text{Sym}^2 \mathbb{R}^n$ are smooth functions on the parameter space $\mathbb{R}^n \times \text{Sym}^2 \mathbb{R}^n$. Compact support allows differentiation under each defining integral to every order. The covariance is positive definite at each finite parameter value. Indeed, the density relative to μ_0 is strictly positive. A nonzero linear functional with zero variance under $\mu_{\theta, \Lambda}$ would then be constant under the isotropic law μ_0 .

Starting from $\theta_0 = 0$ and $\Lambda_0 = 0$, solve locally for the adapted processes $\theta_t \in \mathbb{R}^n$ and $\Lambda_t \in \text{Sym}^2 \mathbb{R}^n$:

$$d\theta_t = A_t^{-1/2} dB_t + A_t^{-1} m_t dt, \quad d\Lambda_t = A_t^{-1} dt, \quad \mu_t = \mu_{\theta_t, \Lambda_t}, \quad (6.1)$$

where $m_t = m(\theta_t, \Lambda_t)$ and $A_t = A(\theta_t, \Lambda_t)$. Here, B_t is standard Brownian motion in $\mathbb{R}^n$, and $\mathcal{F}_t$ is its filtration, with the usual completion and right-continuity. Smoothness of these coefficients gives a unique solution up to its maximal lifetime. We prove below that this lifetime is infinite. Before that time, $\Lambda_t = \int_0^t A_s^{-1} ds$ and $\Lambda_t \succeq 0$, so μ_t is log-concave and has curvature at least Λ_t .

Write $Z_t \in (0, \infty)$ for the normalizing integral and $\mathbb{E}_t$ for integration under the random law μ_t . In this section and the localization arguments below, $\mathbb{E}$ without a subscript averages over the Brownian path. At (θ_t, Λ_t) , differentiation of the normalizing integral gives

$$\nabla_\theta \log Z = m_t, \quad \nabla_\theta^2 \log Z = A_t, \quad D_\Lambda \log Z[H] = -\frac{1}{2} \mathbb{E}_t[X^T H X]$$

for every symmetric matrix direction $H \in \text{Sym}^2 \mathbb{R}^n$. The process Λ_t has finite variation, so its quadratic covariations vanish, while $d[\theta_i, \theta_j]_t = (A_t^{-1})_{ij} dt$. Thus, the three contributions in Itô's formula are

$$\begin{aligned} m_t^T d\theta_t &= m_t^T A_t^{-1/2} dB_t + m_t^T A_t^{-1} m_t dt, \\ D_\Lambda \log Z[d\Lambda_t] &= -\frac{1}{2} \mathbb{E}_t[X^T A_t^{-1} X] dt, \\ \frac{1}{2} \sum_{i,j} (A_t)_{ij} d[\theta_i, \theta_j]_t &= \frac{1}{2} \text{Tr}(A_t A_t^{-1}) dt. \end{aligned}$$

Since $\mathbb{E}_t[XX^T] = A_t + m_t m_t^T$, their drift terms combine as

$$\begin{aligned} m_t^T A_t^{-1} m_t - \frac{1}{2} \mathbb{E}_t[X^T A_t^{-1} X] + \frac{1}{2} \text{Tr}(A_t A_t^{-1}) \\ = m_t^T A_t^{-1} m_t - \frac{1}{2} [\text{Tr}(A_t A_t^{-1}) + m_t^T A_t^{-1} m_t] + \frac{1}{2} \text{Tr}(A_t A_t^{-1}) \\ = \frac{1}{2} m_t^T A_t^{-1} m_t. \end{aligned}$$

Consequently,

$$d \log Z_t = m_t^T A_t^{-1/2} dB_t + \frac{1}{2} m_t^T A_t^{-1} m_t dt.$$

Define $\xi_t : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $\xi_t(x) = A_t^{-1/2}(x - m_t)$; these coordinates give mean zero and identity covariance under μ_t . Let $\rho_t : \mathbb{R}^n \rightarrow (0, \infty)$ be the density $d\mu_t/d\mu_0$. Subtracting this differential from that of $\theta_t \cdot x - x^T \Lambda_t x/2$ yields $d \log \rho_t(x) = \xi_t(x) \cdot dB_t - \frac{1}{2} |\xi_t(x)|^2 dt$. Applying Itô's formula once more gives

$$d\rho_t(x) = \rho_t(x) \xi_t(x) \cdot dB_t, \quad \mu_t(dx) = \rho_t(x) \mu_0(dx). \quad (6.2)$$

Integrating a bounded Borel function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ against this identity, or applying Itô's formula to its normalized tilt integral, gives

$$d\mathbb{E}_t f = \mathbb{E}_t[(f - \mathbb{E}_t f) \xi_t] \cdot dB_t. \quad (6.3)$$

For now this is a local martingale identity; boundedness of the test function will make it a true martingale after global existence is established.

The covariance equation involves third moments. We bound them using Letwin's inequality in Equation (2.8). We keep the tensor factors ordered because they will contain covariance matrices from different times.

First, let Y be any isotropic log-concave random vector in $\mathbb{R}^n$. For this calculation, expectations refer to its law. Set $S_i = \mathbb{E}[Y_i Y Y^T] \in \text{Sym}^2 \mathbb{R}^n$; its entries are third moments with one index fixed. For $v \in \mathbb{R}^n$, put $S(v) = \mathbb{E}[(v \cdot Y) Y Y^T] = \sum_i v_i S_i \in \text{Sym}^2 \mathbb{R}^n$. If $B \in \text{Sym}^2 \mathbb{R}^n$, then $\mathbb{E}(v \cdot Y) = 0$ and $\mathbb{E}(Y^T B Y) = \text{Tr } B$, so

$$|\langle S(v), B \rangle_{\text{HS}}| = |\mathbb{E}[(v \cdot Y)(Y^T B Y - \text{Tr } B)]| \leq |v| \sqrt{\text{Var}(Y^T B Y)} \leq \sqrt{8} |v| \|B\|_{\text{HS}}.$$

The matrix $S(v)$ is symmetric. If it is nonzero, taking $B = S(v) / \|S(v)\|_{\text{HS}}$ in this inequality gives $\|S(v)\|_{\text{HS}}^2 \leq 8|v|^2$; the same conclusion holds when $S(v) = 0$. Symmetry of the three indices in $\mathbb{E}[Y_i Y_j Y_k]$ gives

$$\left\langle v, \left(\sum_i S_i^2 \right) v \right\rangle = \sum_{i,j} \left(\sum_k v_k \mathbb{E}[Y_i Y_j Y_k] \right)^2 = \|S(v)\|_{\text{HS}}^2.$$

Consequently $\sum_i S_i^2 \preceq 8\text{Id}$. Each S_i is self-adjoint, and hence

$$0 \preceq \sum_i (S_i \otimes \text{Id} - \text{Id} \otimes S_i)^2 = \left(\sum_i S_i^2 \right) \otimes \text{Id} + \text{Id} \otimes \left(\sum_i S_i^2 \right) - 2 \sum_i S_i \otimes S_i.$$

It follows that $\sum_i S_i \otimes S_i \preceq 8\text{Id}$ on the entire space $\mathbb{R}^n \otimes \mathbb{R}^n$.

Apply these conclusions conditionally to the isotropic vector $\xi_t = A_t^{-1/2}(X - m_t)$ under μ_t , and define

$$S_{i,t} = \mathbb{E}_t[\xi_{i,t} \xi_t \xi_t^T], \quad U_{i,t} = A_t^{1/2} S_{i,t} A_t^{1/2}.$$

Both $S_{i,t}$ and $U_{i,t}$ are adapted processes in $\text{Sym}^2 \mathbb{R}^n$. Conjugating the tensor inequality by $A_t^{1/2} \otimes A_t^{1/2}$ gives

$$\sum_i S_{i,t}^2 \preceq 8\text{Id}, \quad \sum_i U_{i,t} \otimes U_{i,t} \preceq 8A_t \otimes A_t. \quad (6.4)$$

The mean and covariance equations follow from Equation (6.3). Indeed, applying it to the coordinate functions gives $dm_t = \mathbb{E}_t[(X - m_t) \xi_t^T] dB_t = A_t^{1/2} dB_t$. Apply it also to the entries of $\mathbb{E}_t X X^T$ and differentiate $A_t = \mathbb{E}_t X X^T - m_t m_t^T$. The quadratic variation of m_t contributes $-A_t dt$. The coefficient of $dB_{i,t}$ is

$$\mathbb{E}_t[X X^T \xi_{i,t}] - (A_t^{1/2} e_i) m_t^T - m_t (A_t^{1/2} e_i)^T = \mathbb{E}_t[(X - m_t)(X - m_t)^T \xi_{i,t}] = U_{i,t}.$$

Thus,

$$dm_t = A_t^{1/2} dB_t, \quad dA_t = \sum_i U_{i,t} dB_{i,t} - A_t dt.$$

The next lemma ensures that localization can run for any finite time. Averaged over the random process, the integral of a bounded test function stays equal to its initial value.

Lemma 6.1 (Global Existence). *The solution of Equation (6.1) exists at every finite time. For every bounded Borel function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the process $\mathbb{E}_t f$ is a true martingale.*

The proof is given in Appendix D.1.

We can now estimate covariance moments at arbitrary finite times. For an integer $k \geq 1$, the drift of $A_t^{\otimes k}$ has k terms $-A_t^{\otimes k}$ and one quadratic-variation term for each unordered pair of distinct

slots. In a given pair of slots, that term inserts $\sum_i U_{i,t} \otimes U_{i,t}$ and leaves A_t in every other slot. Tensoring Equation (6.4) with those positive matrices bounds this term above by $8A_t^{\otimes k}$. The drift is therefore at most

$$\left(-k + 8\binom{k}{2}\right) A_t^{\otimes k} = (4k^2 - 5k)A_t^{\otimes k} \preceq 4k^2 A_t^{\otimes k}.$$

To take expectations, we must check that the stochastic integrals are true martingales. Every posterior remains in the original ball of radius R , so $A_t \preceq R^2 \text{Id}$. Also,

$$\sum_i \|U_{i,t}\|_{\text{HS}}^2 \leq R^4 \sum_i \|S_{i,t}\|_{\text{HS}}^2 = R^4 \text{Tr}\left(\sum_i S_{i,t}^2\right) \leq 8nR^4.$$

Write $H_{i,t} : (\mathbb{R}^n)^{\otimes k} \rightarrow (\mathbb{R}^n)^{\otimes k}$ for the linear operator that is the coefficient of $dB_{i,t}$ in $d(A_t^{\otimes k})$. It is the sum of k tensor products, each with one factor $U_{i,t}$ and $k-1$ factors A_t . Therefore, for any fixed $z \in (\mathbb{R}^n)^{\otimes k}$,

$$\sum_i |\langle z, H_{i,t} z \rangle|^2 \leq k^2 R^{4(k-1)} |z|^4 \sum_i \|U_{i,t}\|_{\text{op}}^2 \leq 8nk^2 R^{4k} |z|^4.$$

This deterministic bound is integrable on every finite time interval. The scalar martingale terms are consequently square-integrable. Multiplying the tensor equation by $e^{-4k^2 t}$ makes its drift negative semidefinite. Conditional expectation of its quadratic form on z between times s and t gives

$$\mathbb{E}[A_t^{\otimes k} \mid \mathcal{F}_s] \preceq e^{4k^2(t-s)} A_s^{\otimes k}, \quad 0 \leq s \leq t. \quad (6.5)$$

One may first take z with rational coordinates and then use continuity of quadratic forms, so that the inequality holds simultaneously for all z .

We also require factors evaluated at different times. For arbitrary deterministic $s_1, \dots, s_d \in [0, t]$, we claim that

$$\mathbb{E}[A_{s_1} \otimes \dots \otimes A_{s_d}] \preceq e^{4d^2 t} \text{Id}. \quad (6.6)$$

Permute the slots to put their times in nondecreasing order. Let u be the largest time and k its multiplicity. Let v be the next smaller time, or 0 if all times equal u . The other factors form a positive semidefinite $\mathcal{F}_v$ -measurable matrix G_v on $(\mathbb{R}^n)^{\otimes(d-k)}$; put $G_v = 1$ if there are no other factors. Conditioning entry by entry and using Equation (6.5) gives

$$\begin{aligned} \mathbb{E}[G_v \otimes A_u^{\otimes k}] &= \mathbb{E}\left[G_v \otimes \mathbb{E}[A_u^{\otimes k} \mid \mathcal{F}_v]\right] \\ &\preceq e^{4k^2(u-v)} \mathbb{E}[G_v \otimes A_v^{\otimes k}]. \end{aligned}$$

The order comparison holds because tensoring a positive operator with $G_v \succeq 0$ preserves positivity. Boundedness of the covariance matrices justifies the conditional expectations. This moves all k factors at u to v , at a cost of $e^{4k^2(u-v)}$. Merge them with any factors already at v and repeat. This procedure ends with all factors at time zero, where $A_0 = \text{Id}$. Its successive time intervals are disjoint, have total length at most t , and use multiplicities at most d . The product of the costs is thus at most $e^{4d^2 t}$. Conjugating back by the initial tensor permutation proves Equation (6.6).

The terminal change of coordinates also involves the inverse of the accumulated curvature. A matrix inequality reduces its control to the preceding estimates for mixed covariance moments. For $t > 0$, integrate the positive block matrix acting on $\mathbb{R}^n \oplus \mathbb{R}^n$:

$$\begin{pmatrix} A_s & \text{Id} \\ \text{Id} & A_s^{-1} \end{pmatrix} = \begin{pmatrix} A_s^{1/2} \\ A_s^{-1/2} \end{pmatrix} \begin{pmatrix} A_s^{1/2} & A_s^{-1/2} \end{pmatrix} \succeq 0.$$

The resulting block matrix has diagonal entries $\int_0^t A_s ds$ and Λ_t , and off-diagonal entries $t\text{Id}$. Testing it on $(z, -t\Lambda_t^{-1}z)$, for $z \in \mathbb{R}^n$, shows that

$$\Lambda_t^{-1} \preceq t^{-2} \int_0^t A_s ds.$$

For any j selected slots among d , insert this bound in each selected slot and put A_t in the others. Tensor products preserve the order when all remaining factors are positive semidefinite. After expanding the integrals, Equation (6.6) bounds the expectation of every integrand. The integration region has volume t^j , and consequently, up to the selected permutation of slots,

$$\mathbb{E}[(\Lambda_t^{-1})^{\otimes j} \otimes A_t^{\otimes(d-j)}] \preceq t^{-j} e^{4d^2 t} \text{Id}.$$

The interchange of expectation and integration is justified by the bound $A_s \preceq R^2 \text{Id}$ on the integrands. For an integer $0 \leq q \leq d$, expanding each of the first q factors of $A_t + \delta \Lambda_t^{-1}$ now gives

$$\mathbb{E}[(A_t + \delta \Lambda_t^{-1})^{\otimes q} \otimes A_t^{\otimes(d-q)}] \preceq (1 + \delta/t)^q e^{4d^2 t} \text{Id}, \quad \delta \geq 0. \quad (6.7)$$

There are $\binom{q}{j}$ terms with exactly j selected inverse covariance slots, and their coefficients sum to $\sum_{j=0}^q \binom{q}{j} (\delta/t)^j = (1 + \delta/t)^q$. This proves the mixed estimate on the full ordered tensor space.

6.2 The Moving Appell Variance

Localization changes both the measure and its Appell polynomial. We track their combined effect on the variance. This will let us transfer an estimate for the localized measure back to the original law.

Fix an integer $d \geq 2$ and a tensor $T \in \text{Sym}^d \mathbb{R}^n$, and set

$$p_t = P_d^{\mu_t}[T], \quad v_t = \mathbb{E}_t p_t^2, \quad \mathcal{V}(t) = \mathbb{E} v_t.$$

Here, $p_t : \mathbb{R}^n \rightarrow \mathbb{R}$ is a random polynomial, v_t is a nonnegative real random variable, and $\mathcal{V} : [0, \infty) \rightarrow [0, \infty)$ is its deterministic expectation. The leading coefficient tensor stays fixed in the original coordinates, while the lower-degree coefficients change with the posterior. Because we use the Appell polynomial of the current law, the expected spatial derivatives of orders below d vanish at every time. Each coefficient of p_t is a polynomial in finitely many bounded moment martingales, by Equations (2.2) and (6.3). Thus, Itô's formula applies coefficient by coefficient.

For chosen finite upper bounds $C_1, \dots, C_{d-1} \in [0, \infty)$ on the coefficients of isotropic log-concave laws, define the drift sum

$$\Sigma_d = \sum_{k=2}^{d-1} (d-k+1) C_k C_{d-k+1}, \quad \Sigma_2 = 0. \quad (6.8)$$

The following lemma bounds the original polynomial norm using its norm at a later time. Its error involves only lower polynomial degrees, which makes it suitable for induction. It is proved in Appendix D, which calculates the drift and checks the integrability needed to take expectations.

Lemma 6.2 (Moving Appell Variance). *Fix an integer $d \geq 2$. Let μ_t be the covariance-adapted localization of a compactly supported isotropic log-concave law. Suppose finite coefficient bounds $c_k(\nu) \leq C_k$ hold for every isotropic log-concave law ν and $1 \leq k < d$. With Σ_d defined by Equation (6.8),*

$$\sqrt{\mathcal{V}(0)} \leq e^{(d+1)t} \sqrt{\mathcal{V}(t)} + d! \Sigma_d t e^{(2d^2+d+1)t} \|T\|, \quad t \geq 0. \quad (6.9)$$

The error does not depend on how many derivatives we use to estimate the polynomial at the terminal time.

7 Reverse Coefficient Transfer

In this section, we turn integration bounds into coefficient bounds. We recover the terminal Appell polynomial from its q th derivative using q centered integrations. The derivative has lower degree, and the bound for the whole product of integrations is paid once. The mixed covariance estimate in Equation (6.7) controls the change of coordinates, while Equation (6.9) transfers the terminal estimate back to time zero.

The proposition below gives the estimate needed to advance the induction: it bounds degree d using repeated integration and coefficients of smaller degree.

Proposition 7.1 (Reverse Coefficient Transfer). *Fix integers $d \geq 2$ and $1 \leq q \leq d - 1$, and let $0 < \eta \leq 1$ and $\delta = \eta^2/d^2$. Suppose $M_q < \infty$ is a nonnegative constant such that every centered regular log-concave law ν with $\text{Cov}(\nu) \preceq \text{Id}$ and curvature at least δId satisfies*

$$\left\| J_0^\nu J_1^\nu \cdots J_{q-1}^\nu \right\|_{\text{op}} \leq M_q. \quad (7.1)$$

Suppose finite constants $C_1, \dots, C_{d-1} \geq 0$ satisfy $c_k(\nu) \leq C_k$ for every isotropic log-concave law and $1 \leq k < d$. With Σ_d as in Equation (6.8), the uniform coefficient c_d^ satisfies*

$$c_d^* \leq e^{3\eta} \left[(1 + \eta)^{q/2} M_q C_{d-q} + \frac{\eta}{d^2} \Sigma_d \right]. \quad (7.2)$$

Here, M_q is the operator-norm bound for q integrations. The integer d is the degree to be estimated, and we use q derivatives in the terminal estimate. The parameter η sets the localization time $t = \eta/d^2$. The factor $(1 + \eta)^{q/2}$ is the cost of comparing the q derivative slots with the covariance metric.

7.1 The Terminal Estimate in the Covariance Metric

Proof of Proposition 7.1. We first recover a polynomial from a lower-degree derivative by centered integration. A change of coordinates controls the cost of this step for the localized measure. Averaging and the moving variance estimate then give the bound for the original law.

Let $p : \mathbb{R}^n \rightarrow \mathbb{R}$ be a polynomial for a regular law ν , with

$$\mathbb{E}_\nu D^j p = 0 \quad (0 \leq j \leq q).$$

Its derivatives $D^j p : \mathbb{R}^n \rightarrow \text{Sym}^j \mathbb{R}^n$ are compatible symmetric fields. The centered primitive at each step therefore recovers the preceding derivative, so that

$$p = J_0^\nu J_1^\nu \cdots J_{q-1}^\nu D^q p.$$

Under the covariance and curvature assumptions of the proposition,

$$\mathbb{E}_\nu p^2 \leq M_q^2 \mathbb{E}_\nu \|D^q p\|^2. \quad (7.3)$$

The final constraint $\mathbb{E}_\nu D^q p = 0$ is essential for applying this product. A degree- d Appell polynomial satisfies all the constraints when $q \leq d - 1$; at $q = d$ the final field is constant.

Before applying this estimate to localized Appell polynomials, we extend it to nonsmooth laws. Let ν be a possibly nonsmooth, full-dimensional centered law with $\text{Cov}(\nu) \preceq \text{Id}$ and curvature at least δId , where $0 < \delta \leq 1$. Lemma F.1 gives regular approximants ν_ε with the same covariance and curvature bounds. Every fixed moment also converges. Apply Equation (7.3) to each approximant's degree- d Appell polynomial with a fixed leading tensor. The derivative constraints hold exactly, and Equation (2.7) gives convergence of both integrals. Passing to the limit gives the same bound for ν , with the same number M_q .

We next choose coordinates in which both the covariance and curvature assumptions hold. Let μ be a probability law on $\mathbb{R}^n$ with mean $m \in \mathbb{R}^n$, covariance $A \in \text{Sym}^2 \mathbb{R}^n$, and curvature matrix $B \in \text{Sym}^2 \mathbb{R}^n$, where $A, B \succ 0$. For an $\mathbb{R}^n$ -valued random vector X with law μ , set

$$\mathbf{M} = A + \delta B^{-1}, \quad \nu = \mathcal{L}(\mathbf{M}^{-1/2}(X - m)). \quad (7.4)$$

The positive matrix $\mathbf{M} \in \text{Sym}^2 \mathbb{R}^n$ defines the norm used for derivatives, and ν is the transformed probability law on $\mathbb{R}^n$. The transformed covariance is at most Id . Inverting $\mathbf{M} \succeq \delta B^{-1}$ gives $B \succeq \delta \mathbf{M}^{-1}$; multiplying on the left and right by $\mathbf{M}^{1/2}$ therefore gives $\mathbf{M}^{1/2} B \mathbf{M}^{1/2} \succeq \delta \text{Id}$. The affine transformation rule in Equation (2.6) preserves the derivative constraints. To track the derivative norm, put $\bar{p}(y) = p(m + \mathbf{M}^{1/2}y)$. The chain rule acts separately in each of the q derivative slots:

$$D_y^q \bar{p}(y) = (\mathbf{M}^{1/2})^{\otimes q} D_x^q p(m + \mathbf{M}^{1/2}y).$$

Taking the squared norm multiplies these factors by their transposes and gives $\mathbf{M}^{\otimes q}$. Applying Equation (7.3) to $\bar{p}$ under ν therefore gives, for a degree- d Appell polynomial,

$$\mathbb{E}_\mu p^2 \leq M_q^2 \mathbb{E}_\mu \langle D^q p, \mathbf{M}^{\otimes q} D^q p \rangle, \quad 0 < \delta \leq 1, \quad q \leq d-1. \quad (7.5)$$

Thus, the same matrix controls both the curvature after the change of coordinates and the norm of the derivative.

For the localized Appell polynomial and any positive semidefinite $\mathbf{M} \in \text{Sym}^2 \mathbb{R}^n$, the derivative identity and the degree- $(d-q)$ coefficient bound give

$$\mathbb{E}_t \langle D^q p_t, \mathbf{M}^{\otimes q} D^q p_t \rangle \leq (d!)^2 C_{d-q}^2 \langle T, (\mathbf{M}^{\otimes q} \otimes A_t^{\otimes(d-q)}) T \rangle. \quad (7.6)$$

To verify the normalization, taking q derivatives contributes $d!/(d-q)!$, and the coefficient bound for each scalar component contributes $(d-q)!C_{d-q}$. Expressing the polynomial in isotropic coordinates puts a factor $A_t^{1/2}$ in each of the remaining $d-q$ tensor slots. Apply $\mathbf{M}^{1/2}$ in each derivative slot. More explicitly, put

$$S = ((\mathbf{M}^{1/2})^{\otimes q} \otimes (A_t^{1/2})^{\otimes(d-q)}) T.$$

This tensor belongs to $(\mathbb{R}^n)^{\otimes d}$; the two groups of slots need not be symmetric with each other. For every ordered derivative-index tuple I , the remaining slice $S_I \in \text{Sym}^{d-q} \mathbb{R}^n$ is symmetric. Apply the scalar coefficient bound to that slice and sum over I . The factorials give $d!$, and

$$\sum_I \|S_I\|^2 = \|S\|^2 = \langle T, (\mathbf{M}^{\otimes q} \otimes A_t^{\otimes(d-q)}) T \rangle.$$

Symmetry within the polynomial slots justifies the coefficient bound. The ordered-index sum is exactly the squared tensor norm, even though the derivative and polynomial slots use different metrics.

At time $t > 0$ use Equation (7.5) with $B = \Lambda_t$ and $\mathbf{M} = A_t + \delta\Lambda_t^{-1}$. Combine Equation (7.6) with Equation (6.7) to obtain

$$\mathcal{V}(t) \leq (d!)^2 C_{d-q}^2 M_q^2 (1 + \delta/t)^q e^{4d^2 t} \|T\|^2.$$

For $T \neq 0$, take square roots and insert this bound into Equation (6.9). Dividing by $d! \|T\|$ gives

$$\frac{\sqrt{\mathcal{V}(0)}}{d! \|T\|} \leq e^{(2d^2+d+1)t} \left[(1 + \delta/t)^{q/2} M_q C_{d-q} + t \Sigma_d \right].$$

The factor $e^{2d^2 t}$ comes from the covariance moment estimate; the factor $e^{(d+1)t}$ comes from the moving variance estimate. Set $t = \eta/d^2$, so that the prescribed curvature threshold is $\delta = \eta t \leq 1$. Now $\delta/t = \eta$ and $(2d^2 + d + 1)t \leq 3\eta$ for $d \geq 2$. Taking the supremum over nonzero T proves Equation (7.2) for compactly supported isotropic laws.

For an arbitrary isotropic log-concave random vector $X \in \mathbb{R}^n$, let $X_R \in \mathbb{R}^n$ have its conditional law on $\{|X| \leq R\}$, for $R > 0$. For large R this event has positive probability, and conditioning on the convex ball preserves log-concavity. For every polynomial $h : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$\mathbb{E}h(X_R) = \frac{\mathbb{E}[h(X)\mathbf{1}_{\{|X| \leq R\}}]}{\Pr\{|X| \leq R\}} \longrightarrow \mathbb{E}h(X)$$

by dominated convergence. Write $m_R = \mathbb{E}X_R \in \mathbb{R}^n$ and $A_R = \text{Cov}(X_R) \in \text{Sym}^2 \mathbb{R}^n$. Then $m_R \rightarrow 0$ and $A_R \rightarrow \text{Id}$, so $Y_R = A_R^{-1/2}(X_R - m_R) \in \mathbb{R}^n$ is defined for all sufficiently large R . It is isotropic, log-concave, and compactly supported. Every fixed moment of Y_R converges to the corresponding moment of X , by expanding the polynomial after this affine substitution.

By Equation (2.7), the bound for the Appell coefficients of Y_R passes to X . Hence, Equation (7.2) holds for every isotropic log-concave law.

Finally, a centered law with covariance $A \preceq \text{Id}$ is a linear image, by a contraction, of an isotropic law on its linear support. Equation (2.6) and $\|(A^{1/2})^{\otimes d}\|_{\text{op}} \leq 1$ extend the result to this law. A point mass has zero positive-degree Appell variances. $\square$

8 Uniform Appell Coefficient Bounds

In this section, we combine Lemma 5.1 with Proposition 7.1 to bound the uniform coefficients c_d^* from Equation (2.4). A bound for $\|J_\nu^q\|$ supplies the number M_q in reverse transfer, because the product $J_0^\nu \cdots J_{q-1}^\nu$ is one of its blocks.

We prove a single bound by induction on degree. Up to a fixed cutoff, we use one integration and the quadratic seed in Corollary 5.3. Above the cutoff, we use about half the derivatives and the uniform power estimate. Both steps apply the same reverse-transfer inequality.

8.1 Induction on Degree

Our target has a fixed exponential growth rate. The polynomial decay in its prefactor lets the power lemma give a strictly smaller rate for repeated integration.

Fix the universal radius $R_* > 1$ chosen in Appendix E, which also records the cutoff and the other constants used in the induction.

Define the weight $w : \mathbb{N}_0 \rightarrow (0, 1]$ by $w(j) = (j + 1)^{-4}$. For each integer $d \geq 1$, define

$$\beta_d = R_*^d w(d) = \frac{R_*^d}{(d + 1)^4}. \quad (8.1)$$

The next theorem gives a single exponential growth bound for the normalized Appell polynomial norms. It holds for every degree, measure, and dimension in the definition of c_d^* .

Theorem 8.1 (Uniform Coefficient Growth). *There is a universal $R_* > 1$ such that*

$$c_d^* \leq \beta_d = \frac{R_*^d}{(d + 1)^4} \quad (d \geq 1).$$

Appendix G records an admissible value of R_* and the resulting Poincaré constant.

Closing the Induction. We compare the gain from lower-degree bounds with the costs of integration and localization. For large d , take $q = \lfloor d/2 \rfloor$ derivatives and use the coefficient bounds only through $D = \lfloor \sqrt{d} \rfloor$. Apply Lemma 5.1 with

$$B = R_*^2 e^{-1/(D+1)} < R_*^2.$$

Choose the localization parameter so that the curvature threshold δ is a sufficiently small fixed multiple of $R_*^{-2} d^{-3}$. The last observation in the power lemma has squared norm at most $R_*^{2D} (D + 1)^{-8}$. Since $(D + 1)^8$ is of order d^4 , the strict condition $\delta B^{D+1} > \gamma_D^2$ holds for large d . The remaining observations give a prefactor smaller than two, so

$$\|J_\nu^q\| \leq \sqrt{2} R_*^q \exp\left[-\frac{q}{2(D+1)}\right].$$

After cancellation of the factors R_*^d , the exponential gain is $\exp[-(1/4 + o(1))\sqrt{d}]$. It absorbs the localization cost and the bounded ratio of weights. For smaller degrees, one integration has cost of order $\sqrt{R_*}$, while the ratio β_{d-1}/β_d is of order R_*^{-1} . Their product is small as well. The weight convolution controls the drift in both cases.

Appendix E makes these comparisons quantitative and verifies the induction for a single choice of constants in all degrees.

8.2 Uniform Bounds for Regular Measures

We can now remove the curvature dependence from the Poincaré bound. The coefficient estimate holds in every degree, so we let the cutoff degree tend to infinity in the power estimate. This gives the following corollary.

Corollary 8.2. *Every regular centered log-concave law ν with covariance at most Id and $D^2 V \succeq a \text{Id}$ for some $a > 0$ satisfies*

$$C_P(\nu) \leq 1 + 2R_*^2.$$

The bound is independent of the dimension and of a .

Proof. Theorem 8.1 gives $c_k^* \leq R_*^k / (k + 1)^4$ in every degree. Fix ν and its positive curvature constant a . For each $D \geq 2$, apply Corollary 5.2 with $\rho = R_*$ and $q = 1$.

$$\|J_\nu\|_{\text{op}}^2 \leq 2R_*^2 \max\{1, a^{-1/(D+1)}\}.$$

Letting $D \rightarrow \infty$ gives $\|J_\nu\|_{\text{op}}^2 \leq 2R_*^2$. Equation (4.10) then gives the asserted bound. $\square$

9 Proof of Theorem 1.1

We can now prove Theorem 1.1 and Corollary 1.2. Set $K_P = 1 + 2R_*^2$. By Corollary 8.2, every regular isotropic law satisfies $C_P \leq K_P$. This bound is independent of both Hessian bounds, so the same constant passes to the limit.

Proof of Theorem 1.1. Let μ be an isotropic log-concave probability measure on $\mathbb{R}^n$. By Lemma F.2, there are regular isotropic measures μ_j converging weakly to μ . Their lower Hessian bounds may tend to zero and their upper bounds may diverge; neither affects the constant K_P . For every $f \in C_c^\infty(\mathbb{R}^n)$,

$$\mathrm{Var}_{\mu_j} f \leq K_P \int |\nabla f|^2 d\mu_j.$$

The functions f , f^2 , and $|\nabla f|^2$ are bounded and continuous, so weak convergence gives

$$\mathrm{Var}_\mu f \leq K_P \int |\nabla f|^2 d\mu.$$

A full-dimensional log-concave law is absolutely continuous. Lemma F.3 therefore extends this inequality to every locally Lipschitz $f : \mathbb{R}^n \rightarrow \mathbb{R}$ of finite Dirichlet energy, including the conclusion $f \in L^2(\mu)$. Thus, $C_P(\mu) \leq K_P$.

For a law with covariance $\Sigma \in \mathrm{Sym}^2 \mathbb{R}^n$, $\Sigma \preceq \mathrm{Id}$, translation leaves the Poincaré constant unchanged, so first center the law and work on its linear support. If that support has positive dimension, write $X = \Sigma^{1/2}Y$ with Y isotropic there. For a locally Lipschitz f , apply the isotropic inequality to $g(y) = f(\Sigma^{1/2}y)$. The chain rule and $\|\Sigma\|_{\mathrm{op}} \leq 1$ give

$$\mathrm{Var} f(X) \leq K_P \mathbb{E} |\nabla g(Y)|^2 = K_P \mathbb{E} \langle \nabla f(X), \Sigma \nabla f(X) \rangle \leq K_P \mathbb{E} |\nabla f(X)|^2.$$

For a lower-dimensional support, all gradients in this calculation are intrinsic. For a smooth ambient test function, the intrinsic gradient is the orthogonal projection of its ambient gradient onto the support. Thus, the inequality with the ambient Dirichlet energy follows as well. A point mass has Poincaré constant zero. $\square$

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Appendix

A Proof of the Hodge Estimate

In this appendix, we prove Lemma 3.1 in three steps, using the assumptions and notation of Section 3. First, integration by parts reduces the estimate to the error from projecting weighted divergence onto compatible fields, as in Equation (A.1). Second, the curl of that divergence is algebraic in the tensor field and the Hessian: all derivatives of the field cancel in Equation (A.2). Third, a Hodge projection formula bounds the error by a finite-dimensional curvature calculation. A constrained minimization then shows that the remaining curvature is at least a , independently of the tensor rank.

Proof of Lemma 3.1. The main task is to bound the projection error for weighted divergence. We first work with $F = \nabla^{m+1}\psi$, $\psi \in C_c^\infty(\mathbb{R}^n)$. Proposition B.4 shows that these fields form a graph core for $\mathcal{D}_m^*$. Thus, every field in the domain can be approximated both in norm and in the norm of its adjoint image. This approximation will extend the estimate to the full adjoint domain at the end of the proof.

Step 1: Weighted Divergence and Its Curl. Integration by parts separates the gradient and curvature terms from the projection error. We then compute the curl of the weighted divergence; the cancellation of derivatives makes the later curvature calculation possible.

Put $Y = \operatorname{div}_V F \in C_c^\infty(\mathbb{R}^n; E_m)$. For each ordered m -tuple I , two integrations by parts give, with I empty when $m = 0$,

$$\sum_{i,j} \langle \partial_i^* F_{Ii}, \partial_j^* F_{Ij} \rangle_{L^2} = \sum_{i,j} \mathbb{E}[\partial_j F_{Ii} \partial_i F_{Ij}] + \sum_{i,j} \mathbb{E}[V_{ij} F_{Ii} F_{Ij}].$$

The full derivative of F is symmetric, so the first sum, including I , is $\|\nabla F\|_2^2$. Write $B_K(F) \in [0, \infty)$ for the integrated curvature contribution:

$$B_K(F) = \sum_{I,i,j} \mathbb{E}[V_{ij} F_{Ii} F_{Ij}].$$

Orthogonal projection therefore yields

$$\|P_{\mathcal{C}_m} Y\|_2^2 = \|\nabla F\|_2^2 + B_K(F) - \|(\operatorname{Id} - P_{\mathcal{C}_m})Y\|_2^2. \quad (\text{A.1})$$

When $m = 0$, this is Bochner's identity: $P_{\mathcal{C}_0} = \operatorname{Id}$ and $B_K(F) = \mathbb{E} \langle F, KF \rangle$. The bound $K \succeq a \operatorname{Id}$ proves the core estimate at rank zero. We now assume $m \geq 1$.

The key cancellation is that the curl of Y depends only on the values of F and K :

$$\partial_k Y_{lI} - \partial_l Y_{kI} = \sum_j (V_{kj} F_{lIj} - V_{lj} F_{kIj}), \quad |I| = m - 1. \quad (\text{A.2})$$

Indeed, expand $Y_{lI} = \sum_j (-\partial_j + V_j) F_{lIj}$. Every term differentiating F cancels against its counterpart with k, l interchanged, because $\partial_k F_{lIj} = \partial_l F_{kIj}$. The displayed terms are precisely the derivatives falling on V_j .

Step 2: The Constrained Hodge Complex. We encode the compatibility equations using symmetric tensors and exterior forms. A differential complex is a sequence of differential operators whose successive products vanish. Its Laplacian lets us compute the projection error in Equation (A.1). For $q \in \mathbb{N}_0$, let $e_\alpha \in \text{Sym}^q \mathbb{R}^n$ be the orthonormal basis indexed by multiplicities $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = q$. Its ordered tensor components are $\sqrt{\alpha!/q!}$ on index tuples of multiplicity α and zero elsewhere, where $\alpha! = \prod_i \alpha_i!$. The following operators lower or raise the multiplicity of one index; they are often called annihilation and creation operators. Their restrictions map $a_i : \text{Sym}^q \mathbb{R}^n \rightarrow \text{Sym}^{q-1} \mathbb{R}^n$ for $q \geq 1$ and $a_i^* : \text{Sym}^q \mathbb{R}^n \rightarrow \text{Sym}^{q+1} \mathbb{R}^n$ for $q \geq 0$:

$$a_i e_\alpha = \sqrt{\alpha_i} e_{\alpha - e_i}, \quad a_i^* e_\alpha = \sqrt{\alpha_i + 1} e_{\alpha + e_i}.$$

Here, e_i in $\alpha \pm e_i$ is the i th coordinate unit vector, and the first expression is zero when $\alpha_i = 0$. In particular, a_i annihilates symmetric degree zero. For example, a mixed rank-two basis tensor has entries $1/\sqrt{2}$ at (i, j) and (j, i) , with $i \neq j$, and has norm one. In particular, contraction in one slot of a rank- q tensor is $a_i/\sqrt{q}$. Thus, $[a_i, a_j^*] = \delta_{ij} \text{Id}$, and $\sum_i a_i^* a_i = q \text{Id}$ on symmetric degree q .

For an integer $p \geq 0$, the space $\Lambda^p \mathbb{R}^n$ consists of alternating p -tensors, or exterior forms, with $\Lambda^0 \mathbb{R}^n = \mathbb{R}$ and $\Lambda^p \mathbb{R}^n = \{0\}$ for $p > n$. Wedge multiplication is the map $\varepsilon_i : \Lambda^p \mathbb{R}^n \rightarrow \Lambda^{p+1} \mathbb{R}^n$ for $p \geq 0$, and contraction is $\iota_i : \Lambda^p \mathbb{R}^n \rightarrow \Lambda^{p-1} \mathbb{R}^n$ for $p \geq 1$. Contraction is zero at degree zero and $\iota_i = \varepsilon_i^*$ between adjacent degrees. They satisfy

$$\varepsilon_i \iota_j + \iota_j \varepsilon_i = \delta_{ij} \text{Id}.$$

Set

$$b = \sum_i \varepsilon_i a_i, \quad b^* = \sum_i a_i^* \iota_i.$$

For $q \geq 1$, these are adjoint maps between $\text{Sym}^q \mathbb{R}^n \otimes \Lambda^p \mathbb{R}^n$ and $\text{Sym}^{q-1} \mathbb{R}^n \otimes \Lambda^{p+1} \mathbb{R}^n$; b is zero when $q = 0$. Thus, b transfers one symmetric index to an exterior index. Expansion of the products gives $b^2 = 0$ and

$$bb^* + b^*b = (q + p) \text{Id} \quad \text{on } \text{Sym}^q \mathbb{R}^n \otimes \Lambda^p \mathbb{R}^n. \quad (\text{A.3})$$

In bb^* , commute a_i past a_j^* ; the terms $a_j^* a_i \varepsilon_i \iota_j$ cancel with their counterparts in b^*b after using $\iota_j \varepsilon_i = \delta_{ij} - \varepsilon_i \iota_j$. The remaining operators are $\sum_i a_i^* a_i$ and $\sum_i \varepsilon_i \iota_i$, with eigenvalues q and p .

Fix the symmetric degree $q = m - 1$, and define the spaces of tensor values, also called fibers,

$$\mathcal{E}_p = \ker b \subset \text{Sym}^{m-1} \mathbb{R}^n \otimes \Lambda^p \mathbb{R}^n, \quad p \geq 1.$$

They are fixed, finite-dimensional subspaces, independent of x . A field taking values in one of these spaces is also called a section. Their orthogonal projections $P_p : \text{Sym}^{m-1} \mathbb{R}^n \otimes \Lambda^p \mathbb{R}^n \rightarrow \mathcal{E}_p$ are, by Equation (A.3),

$$P_p = \frac{bb^*}{m + p - 1}. \quad (\text{A.4})$$

To check this, bb^* maps into $\ker b$, and it equals $(m + p - 1) \text{Id}$ on $\ker b$ by Equation (A.3). The map

$$\mathcal{I}_m : \text{Sym}^m \mathbb{R}^n \longrightarrow \mathcal{E}_1, \quad \mathcal{I}_m T = \frac{bT}{\sqrt{m}},$$

is an isometry onto $\mathcal{E}_1$. Its i th exterior component is the one-slot contraction $a_i T / \sqrt{m}$. Indeed, on symmetric degree m and exterior degree zero, Equation (A.3) gives $b^* b T = m T$, so

$$\|\mathcal{I}_m T\|^2 = \frac{1}{m} \langle T, b^* b T \rangle = \|T\|^2.$$

Conversely, if $\omega \in \mathcal{E}_1$, then $b\omega = 0$ and $bb^*\omega = m\omega$. Hence, $\omega = \mathcal{I}_m(b^*\omega/\sqrt{m})$, proving surjectivity.

On smooth tensor-valued forms, let

$$d = \sum_i \varepsilon_i \partial_i$$

be exterior differentiation. Since $bd + db = 0$, its restrictions are closed operators

$$\begin{aligned} d_p : \text{Dom}(d_p) &\subseteq L^2(\mu; \mathcal{E}_p) \longrightarrow L^2(\mu; \mathcal{E}_{p+1}), \\ \text{Dom}(d_p) &= \{\omega \in L^2(\mu; \mathcal{E}_p) : d\omega \in L^2(\mu; \mathcal{E}_{p+1})\}. \end{aligned}$$

The derivatives in this domain definition are distributional, and $d_{p+1}d_p = 0$. The adjoint d_p^* maps its domain in $L^2(\mu; \mathcal{E}_{p+1})$ to $L^2(\mu; \mathcal{E}_p)$ and is the projected weighted derivative $P_p d^*$, where $d^* = \sum_i \iota_i \partial_i^*$ (Lemma B.5). Acting pointwise, $\mathcal{I}_m$ identifies $\mathcal{C}_m$ with $\ker d_1 \subseteq L^2(\mu; \mathcal{E}_1)$; the kernel equations are precisely those in Equation (3.1).

We estimate the projection error through the Laplacian of this complex. Let $\mathcal{L}_2 : \text{Dom}(\mathcal{L}_2) \subseteq L^2(\mu; \mathcal{E}_2) \rightarrow L^2(\mu; \mathcal{E}_2)$ be the nonnegative self-adjoint operator associated with the closed quadratic form

$$\mathfrak{l}_2(\omega) = \|d\omega\|_2^2 + \|P_1 d^* \omega\|_2^2.$$

Its form domain is $\text{Dom}(d_2) \cap \text{Dom}(d_1^*)$. We first work with $\omega \in C_c^\infty(\mathbb{R}^n; \mathcal{E}_2)$. Lemma B.5 approximates a field together with both derivatives, extending the resulting bound to the full form domain. Write

$$K_{\text{fer}} = \sum_{i,j} V_{ij} \varepsilon_i \iota_j, \quad K_{\text{bos}} = \sum_{i,j} V_{ij} a_j^* a_i.$$

At each x , both are linear operators on $\text{Sym}^{m-1} \mathbb{R}^n \otimes \Lambda^2 \mathbb{R}^n$. They apply curvature to the exterior and symmetric indices, respectively. Using $\partial_i \partial_j^* = \partial_j^* \partial_i + V_{ij}$ and $\varepsilon_i \iota_j + \iota_j \varepsilon_i = \delta_{ij} \text{Id}$, we compute on smooth compactly supported forms:

$$\begin{aligned} dd^* + d^*d &= \sum_{i,j} (\varepsilon_i \iota_j \partial_i \partial_j^* + \iota_j \varepsilon_i \partial_j^* \partial_i) \\ &= \sum_{i,j} (\varepsilon_i \iota_j + \iota_j \varepsilon_i) \partial_j^* \partial_i + \sum_{i,j} V_{ij} \varepsilon_i \iota_j \\ &= \sum_i \partial_i^* \partial_i + K_{\text{fer}}. \end{aligned}$$

Taking the $L^2(\mu)$ inner product with ω and integrating by parts gives the weighted Weitzenböck identity

$$\|d\omega\|_2^2 + \|d^* \omega\|_2^2 = \|\nabla \omega\|_2^2 + \mathbb{E} \langle \omega, K_{\text{fer}} \omega \rangle.$$

The field $d^* \omega$ has symmetric degree $m-1$ and exterior degree one. Hence, Equation (A.4) gives $\text{Id} - P_1 = b^* b / m$, and therefore

$$\|(\text{Id} - P_1) d^* \omega\|_2^2 = \frac{1}{m} \|b d^* \omega\|_2^2.$$

Define

$$c = \sum_i a_i \partial_i^*, \quad c^* = \sum_i a_i^* \partial_i.$$

On smooth compactly supported fields, c maps values in $\text{Sym}^q \mathbb{R}^n \otimes \Lambda^p \mathbb{R}^n$ to values in $\text{Sym}^{q-1} \mathbb{R}^n \otimes \Lambda^p \mathbb{R}^n$ for $q \geq 1$. Its formal weighted adjoint c^* reverses this map between adjacent degrees. Thus,

these differential operators change only the symmetric degree; c is zero at degree zero. The maps a_i commute with the exterior maps and the weighted derivatives, so the exterior anticommutation relation gives

$$bd^* + d^*b = \sum_{i,j} a_i \partial_j^* (\varepsilon_i \iota_j + \iota_j \varepsilon_i) = \sum_i a_i \partial_i^* = c.$$

Since $b\omega = 0$, this implies $bd^*\omega = c\omega$. Therefore,

$$\mathfrak{l}_2(\omega) = \|\nabla\omega\|_2^2 + \mathbb{E} \langle \omega, K_{\text{fer}}\omega \rangle - \frac{1}{m} \|c\omega\|_2^2. \quad (\text{A.5})$$

A separate expansion gives

$$[c, c^*] = H - K_{\text{bos}}. \quad (\text{A.6})$$

Commuting a_i past a_j^* in cc^* isolates the diagonal sum. The remaining terms use $[\partial_j, \partial_i^*] = V_{ij}$:

$$\begin{aligned} cc^* - c^*c &= \sum_i \partial_i^* \partial_i + \sum_{i,j} a_j^* a_i (\partial_i^* \partial_j - \partial_j \partial_i^*) \\ &= H - \sum_{i,j} V_{ij} a_j^* a_i. \end{aligned}$$

The row operator $(a_i^*)_i$, which raises symmetric degree, maps $\bigoplus_i \text{Sym}^{m-1} \mathbb{R}^n$ to $\text{Sym}^m \mathbb{R}^n$ and has norm $\sqrt{m}$. Its product with its adjoint is $\sum_i a_i^* a_i = m \text{Id}$ on the target. Applying this row pointwise to $(\partial_i \omega)_i$ gives

$$\|c^*\omega\|_2^2 \leq m \|\nabla\omega\|_2^2.$$

Together with Equation (A.6), this implies

$$\begin{aligned} \|c\omega\|_2^2 &= \|c^*\omega\|_2^2 - \|\nabla\omega\|_2^2 + \mathbb{E} \langle \omega, K_{\text{bos}}\omega \rangle \\ &\leq (m-1) \|\nabla\omega\|_2^2 + \mathbb{E} \langle \omega, K_{\text{bos}}\omega \rangle. \end{aligned}$$

Substitution into Equation (A.5) proves

$$\mathfrak{l}_2(\omega) \geq \frac{1}{m} \|\nabla\omega\|_2^2 + \mathbb{E} \langle \omega, Q_m(K)\omega \rangle, \quad Q_m(K) = P_2 \left(K_{\text{fer}} - \frac{1}{m} K_{\text{bos}} \right) P_2. \quad (\text{A.7})$$

Here, $Q_m(K) : \mathcal{E}_2 \rightarrow \mathcal{E}_2$ is a self-adjoint operator on each fiber. We next compute this curvature operator after projection onto the allowed tensor space. The calculation shows that it is positive despite the minus sign in its definition.

Step 3: The Curvature Calculation. We now compute the curvature term at a fixed point. The calculation gives a bound on the projection loss and shows that subtracting it retains the lower curvature bound.

Fix $x \in \mathbb{R}^n$ and diagonalize $K(x)$, writing its positive eigenvalues as $\kappa_1, \dots, \kappa_n \in (0, \infty)$. The following calculation is pointwise and orthogonally invariant, so the eigenbasis can be chosen separately at each point. Put $N = m + 1$. Both b and $Q_m(K)$ preserve the combined index multiplicities $\alpha \in \mathbb{N}_0^n$, where symmetric and exterior indices together satisfy $\sum_i \alpha_i = N$. In this block an exterior pair $i < j$ has symmetric multiplicity $\alpha - e_i - e_j$. Let $S_\alpha = \{i : \alpha_i > 0\} \subseteq \{1, \dots, n\}$. All vectors and the diagonal matrix K in this block calculation are restricted to the coordinate space $\mathbb{R}^{S_\alpha}$. Set

$$r = (\sqrt{\alpha_i})_{i \in S_\alpha} \in \mathbb{R}^{S_\alpha}, \quad S = \sum_i \alpha_i \kappa_i \in (0, \infty), \quad |r|^2 = N.$$

In the normalized basis, b becomes wedge multiplication by r on this multiplicity block. To see this, apply b to an exterior pair: inserting an index i already in the pair gives zero, while inserting an index outside the pair lowers symmetric multiplicity α_i and contributes $\sqrt{\alpha_i} = r_i$, with the exterior sign. Thus, the condition $b\omega = 0$ becomes $r \wedge \omega = 0$. This identifies the block $\mathcal{E}_{2,\alpha} \subseteq \mathcal{E}_2$ with the two-forms in $\Lambda^2 \mathbb{R}^{S_\alpha}$ satisfying that condition. Every such two-form has a unique representation

$$\omega = r \wedge v, \quad v \in r^\perp \subseteq \mathbb{R}^{S_\alpha}, \quad \|\omega\|^2 = N|v|^2. \quad (\text{A.8})$$

For the representation, let $\iota_r = \sum_i r_i \iota_i$ denote contraction with r . The exterior anticommutation relation gives

$$\iota_r(r \wedge \omega) + r \wedge \iota_r \omega = N\omega.$$

Since $r \wedge \omega = 0$, take $v = N^{-1} \iota_r \omega$. The identity $\iota_r^2 = 0$ gives $v \perp r$, and $|r \wedge v|^2 = |r|^2 |v|^2 - (r \cdot v)^2 = N|v|^2$ proves both the norm formula and uniqueness. If S_α has one element, the two-form block is zero and the following inverse calculations are interpreted in that zero-dimensional space.

On the exterior pair ij , the curvature coefficient before projection is

$$\kappa_i + \kappa_j - \frac{S - \kappa_i - \kappa_j}{m} = \frac{N(\kappa_i + \kappa_j) - S}{m}.$$

For $v \perp r$, expanding the squares yields

$$\sum_{i < j} (\kappa_i + \kappa_j) (r_i v_j - r_j v_i)^2 = S|v|^2 + N \sum_i \kappa_i v_i^2. \quad (\text{A.9})$$

Indeed, the diagonal terms in the expansion sum to $S|v|^2 + N \sum_i \kappa_i v_i^2 - 2 \sum_i \kappa_i r_i^2 v_i^2$. The mixed terms sum to $2 \sum_i \kappa_i r_i^2 v_i^2 - 2(r \cdot v) \sum_i \kappa_i r_i v_i$; the last term vanishes. Combining Equations (A.8) and (A.9) gives

$$\langle \omega, Q_m(K) \omega \rangle = \frac{N^2}{m} \langle v, Kv \rangle. \quad (\text{A.10})$$

In particular,

$$Q_m(K) \succeq \frac{N}{m} a \text{Id} \quad \text{on } \mathcal{E}_2.$$

Thus, $\mathcal{L}_2$ has a positive lower bound and a bounded inverse. More precisely, let $P : \mathbb{R}^{S_\alpha} \rightarrow r^\perp$ be orthogonal projection, and define the isometry

$$U_\alpha : r^\perp \longrightarrow \mathcal{E}_{2,\alpha}, \quad U_\alpha v = \frac{r}{\sqrt{N}} \wedge v.$$

Equation (A.10) becomes

$$U_\alpha^* Q_m(K) U_\alpha = \frac{N}{m} (PKP)|_{r^\perp}. \quad (\text{A.11})$$

We now evaluate the inverse curvature form on the curl of the weighted divergence. Let $F_\alpha \in \mathbb{R}$ be the coefficient of $F(x)$ in the normalized symmetric rank- N basis. Equation (A.2), after the isometry $\mathcal{I}_m$, gives the exterior-pair coefficients of the curl field $d\mathcal{I}_m Y \in C_c^\infty(\mathbb{R}^n; \mathcal{E}_2)$. Write $t \in \mathcal{E}_{2,\alpha}$ for its component in the current block:

$$t_{ij} = \frac{(\kappa_i - \kappa_j) \sqrt{\alpha_i \alpha_j}}{\sqrt{Nm}} F_\alpha. \quad (\text{A.12})$$

To check the normalization, contracting a symmetric rank- N tensor once is $a_j/\sqrt{N}$, and applying $\mathcal{I}_m$ contracts once more by $a_i/\sqrt{m}$. Thus, for $i \neq j$,

$$\frac{a_i}{\sqrt{m}} \frac{a_j}{\sqrt{N}} e_\alpha = \frac{\sqrt{\alpha_i \alpha_j}}{\sqrt{Nm}} e_{\alpha - e_i - e_j}.$$

Multiplication by the curvature difference $\kappa_i - \kappa_j$ in Equation (A.2) gives Equation (A.12). The wedge coefficient is $(r \wedge Kr)_{ij} = (\kappa_j - \kappa_i) \sqrt{\alpha_i \alpha_j}$, which accounts for the minus sign below. Moreover, $(\text{Id} - P)Kr = (S/N)r$, so

$$r \wedge Kr = r \wedge PKr = \sqrt{N} U_\alpha(PKr).$$

It follows that

$$\begin{aligned} t &= -\frac{F_\alpha}{\sqrt{Nm}} r \wedge Kr = -\frac{F_\alpha}{\sqrt{m}} U_\alpha(PKr), \\ U_\alpha^* t &= -\frac{F_\alpha}{\sqrt{m}} PKr, \end{aligned}$$

where the second identity uses $PKr \in r^\perp$ and $U_\alpha^* U_\alpha = \text{Id}$ on that space. Applying Equation (A.11) now gives

$$\langle t, Q_m(K)^{-1} t \rangle = \frac{|F_\alpha|^2}{N} \langle PKr, (PKP|_{r^\perp})^{-1} PKr \rangle. \quad (\text{A.13})$$

The contribution of this component to $B_K(F)$ is $(S/N)|F_\alpha|^2$: on a symmetric tensor, curvature in one slot has the same quadratic form as its average over all N slots. To bound the curvature remaining after projection, minimize $\langle r + w, K(r + w) \rangle$ over $w \in r^\perp$. The cross term is $2\langle w, PKr \rangle$ and the quadratic term is $\langle w, (PKP|_{r^\perp})w \rangle$. Completing the square therefore gives

$$\begin{aligned} S - \langle PKr, (PKP|_{r^\perp})^{-1} PKr \rangle &= \min_{w \perp r} \langle r + w, K(r + w) \rangle \\ &\geq a \min_{w \perp r} |r + w|^2 = aN. \end{aligned} \quad (\text{A.14})$$

The last equality uses $|r|^2 = N$ and $|r + w|^2 = N + |w|^2$. Combining this with Equation (A.13) yields

$$\langle t, Q_m(K)^{-1} t \rangle \leq \left(\frac{S}{N} - a \right) |F_\alpha|^2. \quad (\text{A.15})$$

Thus, subtracting the projection loss leaves at least $a|F_\alpha|^2$ in every multiplicity block. The estimate is uniform in rank because the block calculation is exact; the nonnegative term $m^{-1} \|\nabla \omega\|_2^2$ in Equation (A.7) may be discarded.

Projection Estimate and Extension to the Adjoint Domain. Combining the projection formula with the curvature calculation proves the estimate for smooth fields. We then pass to all fields in the adjoint domain by approximation.

For the closed complex on $L^2(\mu; \mathcal{E}_1)$, $L^2(\mu; \mathcal{E}_2)$, and $L^2(\mu; \mathcal{E}_3)$, Lemma B.6 and the positive lower bound for $\mathcal{L}_2$ give, for $u \in \text{Dom}(d_1)$,

$$\|(\text{Id} - P_{\ker d})u\|_2^2 = \langle du, \mathcal{L}_2^{-1} du \rangle_{L^2}. \quad (\text{A.16})$$

Apply Equation (A.16) to $u = \mathcal{I}_m Y \in C_c^\infty(\mathbb{R}^n; \mathcal{E}_1)$. The isometry identifies $\ker d$ with $\mathcal{C}_m$. By Equation (A.7), $\mathcal{L}_2 \succeq Q_m(K)$ as positive forms, where the latter denotes pointwise multiplication by the positive fiber operator. The inverse comparison in Equation (B.6) therefore gives

$$\|(\text{Id} - P_{\mathcal{C}_m})Y\|_2^2 \leq \mathbb{E} \langle d\mathcal{I}_m Y, Q_m(K)^{-1} d\mathcal{I}_m Y \rangle.$$

Sum Equation (A.15) over the multiplicity blocks and insert the result into Equation (A.1). This proves

$$\|P_{C_m} Y\|_2^2 \geq \|\nabla F\|_2^2 + a \|F\|_2^2.$$

Equation (3.4) identifies the left side with $\|\mathcal{D}_m^* F\|_2^2$.

It remains to extend the estimate from smooth potential fields to the full adjoint domain. By Proposition B.4, for any $F \in \text{Dom}(\mathcal{D}_m^*)$ there are potential fields F_k with $F_k \rightarrow F$ and $\mathcal{D}_m^* F_k \rightarrow \mathcal{D}_m^* F$ in L^2 . Apply the core estimate to $F_k - F_l$. It shows that ∇F_k is Cauchy in L^2 . Closedness of the distributional derivative gives $F \in W^{1,2}$ and $\nabla F_k \rightarrow \nabla F$. The field norms and adjoint norms also converge, so passing to the limit proves Equation (3.5) on the stated domain, including $m = 0$. $\square$

B Domains, Primitives, and Closure

Here, we justify three steps used in Lemma 3.1. We construct primitives of compatible fields and approximate fields together with their operator images. We then express the compatibility projection through a Laplacian. Throughout, μ is regular as in Definition 2.1, with $K = D^2 V$. Tensor norms use all ordered indices. Sobolev derivatives and compatibility identities are understood in distributions. The weight is smooth and strictly positive, so weighted and ordinary Sobolev spaces agree locally, with equivalent norms on each compact set. Write $p = C_P(\mu) \in (0, \infty)$, so that, for $h \in W^{1,2}(\mu)$,

$$p \leq a^{-1}, \quad \text{Var}_\mu(h) \leq p \mathbb{E}|\nabla h|^2.$$

The ordinary curvature bound $p \leq a^{-1}$ is the Bakry–Émery bound (Bakry et al., 2014, Chapter 4); see also Klartag (2023, Equation (1.7)). This scalar Poincaré inequality suffices to construct the primitives. Cutoff constants in this appendix may depend on the fixed dimension, derivative order, and cutoff profile. They occur only in errors that tend to zero, and do not enter the uniform estimates in the main text.

B.1 Compatible Fields and Their Canonical Primitives

The integration operators require primitives that remain square-integrable. We construct them component by component and use the scalar Poincaré inequality to control their norms.

We use the compatible spaces $\mathcal{C}_r$ and their centered subspaces $\mathcal{G}_r$ from Section 3, including the stated rank-zero conventions. Compatibility is closed under $L^2(\mu)$ convergence, since this convergence implies distributional convergence on compact sets. Constant tensors are compatible and

$$\mathcal{C}_r = \mathcal{G}_r \oplus \text{Sym}^r \mathbb{R}^n$$

orthogonally, where the second summand consists of constant fields.

The following lemma lets us integrate any compatible field. Requiring the result to have mean zero makes it unique, and the scalar Poincaré inequality controls its norm.

Lemma B.1 (Canonical Primitive). *For every integer $r \geq 0$ and $F \in \mathcal{C}_{r+1}$, there is a unique $U \in \mathcal{G}_r \cap W^{1,2}(\mu; E_r)$ such that $\nabla U = F$. The map*

$$\tilde{J}_r : \mathcal{C}_{r+1} \longrightarrow \mathcal{G}_r, \quad \tilde{J}_r F = U,$$

is linear and bounded, and

$$\|\tilde{J}_r F\|_2^2 \leq p \|F\|_2^2. \quad (\text{B.1})$$

Its restriction to $\mathcal{G}_{r+1}$ is the operator J_r used in Section 4.

Proof. A vector field $f \in L_{\text{loc}}^2(\mathbb{R}^n; \mathbb{R}^n)$ with zero distributional curl has a global $W_{\text{loc}}^{1,2}(\mathbb{R}^n)$ primitive, as follows. Convolve f with a compactly supported smooth approximate identity. The resulting field $f_\varepsilon \in C^\infty(\mathbb{R}^n; \mathbb{R}^n)$ still has zero curl, and

$$h_\varepsilon(x) = \int_0^1 \langle f_\varepsilon(tx), x \rangle dt$$

defines $h_\varepsilon \in C^\infty(\mathbb{R}^n)$ with $\nabla h_\varepsilon = f_\varepsilon$. Subtract the Lebesgue mean of h_ε on the unit ball. On each fixed ball containing the unit ball, the gradients converge in L^2 . The mean-zero condition on the unit ball controls the additive constants. The usual Poincaré inequality therefore shows that h_ε is Cauchy in $W^{1,2}$ on each such ball. The local limits agree on overlapping balls and give the desired global primitive.

For $h \in W_{\text{loc}}^{1,2}(\mathbb{R}^n)$ with $\nabla h \in L^2(\mu; \mathbb{R}^n)$, the Sobolev part of Lemma F.3 gives $h \in L^2(\mu)$. That lemma uses only the assumed scalar Poincaré inequality on compact smooth tests and positivity and smoothness of the density. Thus, it is independent of the tensor estimates. Subtracting $\mathbb{E}h$ gives the centered primitive with

$$\mathbb{E}h = 0, \quad \|h\|_2^2 \leq p \|\nabla h\|_2^2. \quad (\text{B.2})$$

Now fix $I \in \{1, \dots, n\}^r$. The vector field $(F_{iI})_i$ has zero curl by compatibility, so the preceding argument gives a unique centered scalar primitive $U_I \in W^{1,2}(\mu)$ with $\partial_i U_I = F_{iI}$. Permuting I leaves this gradient unchanged. Uniqueness of the centered primitive therefore makes U symmetric. If $r \geq 1$, then

$$\partial_i U_{jI} = F_{ijI} = F_{jiI} = \partial_j U_{iI},$$

so U is compatible as well. Summing Equation (B.2) over the tensor components gives Equation (B.1). A centered field with zero gradient is zero, which proves uniqueness and hence linearity. $\square$

B.2 Potential Cores and the Raw Derivative

Integration by parts is easiest for smooth, compactly supported potentials. We show that their derivatives approximate every compatible field, with convergence of the first derivative whenever it is square-integrable.

For an integer $r \geq 0$, define

$$\mathcal{K}_r = \{\nabla^r \psi : \psi \in C_c^\infty(\mathbb{R}^n)\}, \quad \mathcal{K}_r^0 = \{\nabla^r \psi - \mathbb{E} \nabla^r \psi : \psi \in C_c^\infty(\mathbb{R}^n)\}.$$

These are subspaces $\mathcal{K}_r \subset \mathcal{C}_r$ and $\mathcal{K}_r^0 \subset \mathcal{G}_r$. For $r = 0$, the convention is $\nabla^0 \psi = \psi$.

The next lemma gives this approximation for both the full compatible space and its mean-zero subspace.

Lemma B.2 (Potential Cores). *The spaces $\mathcal{K}_r$ and $\mathcal{K}_r^0$ are dense in $\mathcal{C}_r$ and $\mathcal{G}_r$, respectively. They are also graph cores for the corresponding ordinary derivatives on their maximal $W^{1,2}(\mu; E_r)$ domains.*

Proof. For $r \geq 1$ and $F \in \mathcal{C}_r$, iterating Lemma B.1 gives $\phi \in W^{r,2}(\mu)$ with $\nabla^r \phi = F$. For $R > 0$, let $\chi_R \in C_c^\infty(\mathbb{R}^n)$ be equal to one on the ball of radius R , zero outside the ball of radius $2R$, and satisfy $|\nabla^j \chi_R| \leq C_j R^{-j}$. The product rule and integrability of all lower derivatives show

$$\nabla^j(\chi_R \phi) \longrightarrow \nabla^j \phi \quad \text{in } L^2(\mu), \quad 0 \leq j \leq r.$$

Indeed, the term without derivatives on χ_R converges by dominated convergence. Every other term has norm at most a constant times $R^{-k} \|\nabla^{j-k} \phi\|_2$ for some $k \geq 1$. If additionally $F \in W^{1,2}$, then $\nabla^{r+1} \phi = \nabla F \in L^2$, and the same argument applies also to $j = r + 1$.

Mollifying the compactly supported Sobolev function $\chi_R \phi$ gives smooth compactly supported functions. They converge in all these weighted norms because the weight and its reciprocal are bounded on the relevant compact set. A diagonal sequence proves the uncentered assertions. Subtracting expectations is a bounded projection and proves the centered assertions. The case $r = 0$ is the same cutoff and mollification argument applied directly to F . $\square$

The raw derivative $\mathcal{D}_r$ from Equation (3.2) is closed on its maximal $W^{1,2}(\mu)$ domain. Lemma B.2 gives density of that domain, and Lemma B.1 gives the bounded inverse $\mathcal{D}_r^{-1} = \tilde{J}_r$.

B.3 Regularity of the Weak Energy Equation

To identify the adjoint domain, we need to recover a derivative from a weak equation. The next lemma supplies the one derivative not already assumed to be square-integrable.

Lemma B.3 (Weak High-Order Energy Equation). *Let $q \geq 1$ be an integer. Suppose $\phi \in W^{q-1,2}(\mu)$ and $G \in L^2(\mu; (\mathbb{R}^n)^{\otimes q})$. Suppose, for every $\psi \in C_c^\infty(\mathbb{R}^n)$,*

$$\int \langle \nabla^q \phi, \nabla^q \psi \rangle d\mu = \int \langle G, \nabla^q \psi \rangle d\mu, \quad (\text{B.3})$$

where the left side initially denotes the distributional pairing. Then $\phi \in W^{q,2}(\mu)$.

Proof. We first prove local regularity, then use cutoffs to obtain global square integrability. In the local argument, $H^s(\mathbb{R}^n)$, $s \in \mathbb{R}$, denotes the usual unweighted Sobolev space; negative orders are defined by duality. Put $w = Z^{-1}e^{-V} \in C^\infty(\mathbb{R}^n; (0, \infty))$. Equation (B.3) says, in distributions,

$$\sum_I \partial_I (w \partial_I \phi) = \sum_I \partial_I (w G_I),$$

where $I = (i_1, \dots, i_q) \in \{1, \dots, n\}^q$ and $\partial_I = \partial_{i_1} \cdots \partial_{i_q}$. Expand the derivatives falling on w . The top-order term on the left is $w \Delta^q \phi$, and all remaining terms are smooth coefficients times derivatives of ϕ of order at most $2q - 1$. Since $\phi \in H_{\text{loc}}^{q-1}$, each such term belongs to H_{loc}^{-q} . The right side belongs to that same space because G is locally L^2 . Multiplication by $1/w$ is smooth locally; hence,

$$\Delta^q \phi \in H_{\text{loc}}^{-q}.$$

Let $\zeta \in C_c^\infty(\mathbb{R}^n)$. The commutator $[\Delta^q, \zeta] = \Delta^q(\zeta \cdot) - \zeta \Delta^q$ has order at most $2q - 1$. Hence,

$$\Delta^q(\zeta \phi) = \zeta \Delta^q \phi + [\Delta^q, \zeta] \phi \in H^{-q}(\mathbb{R}^n), \quad \zeta \phi \in H^{q-1}(\mathbb{R}^n).$$

Writing $\widehat{\zeta \phi} \in L^2(\mathbb{R}^n; \mathbb{C})$ for the Fourier transform, with frequency variable $\xi \in \mathbb{R}^n$, the first statement says that the following integral is finite:

$$\int_{\mathbb{R}^n} \frac{|\xi|^{4q}}{(1 + |\xi|^2)^q} |\widehat{\zeta \phi}(\xi)|^2 d\xi.$$

On $|\xi| \geq 1$, the weight is comparable to $(1 + |\xi|^2)^q$. On the unit ball, the already finite H^{q-1} norm controls the desired integral. Thus, $\zeta\phi \in H^q$, so $\phi \in H_{\text{loc}}^q$.

Choose $\eta \in C_c^\infty(\mathbb{R}^n)$ supported in the ball of radius two, equal to one on the unit ball, and with $0 \leq \eta \leq 1$. Put $\eta_R(x) = \eta(x/R)$ for $R > 0$. Local regularity and compact Sobolev approximation allow us to test Equation (B.3) with $\psi = \eta_R^{2q}\phi$. The product rule gives

$$\nabla^q(\eta_R^{2q}\phi) = \eta_R^{2q}\nabla^q\phi + R_R, \quad |R_R| \leq C_q\eta_R^q \sum_{j=1}^q R^{-j}|\nabla^{q-j}\phi|.$$

Here, $R_R \in L^2(\mu; \text{Sym}^q \mathbb{R}^n)$ collects the terms with a derivative on the cutoff, and $C_q > 0$ depends only on its profile and on q . To check the power of η_R , view η_R^{2q} as a product of $2q$ copies of η_R . In its j th derivative, at least $2q - j \geq q$ copies remain undifferentiated; the differentiated factors are bounded by a constant times R^{-j} .

Define the nonnegative real numbers

$$A_R = \|\eta_R^q \nabla^q \phi\|_2, \quad b_R = C_q \sum_{j=1}^q R^{-j} \|\nabla^{q-j} \phi\|_2.$$

The remainder estimate gives $\|R_R\|_2 \leq b_R$ and $\|R_R/\eta_R^q\|_2 \leq b_R$, with the quotient defined as zero when $\eta_R = 0$. Substituting into the weak equation and applying Cauchy–Schwarz yields

$$A_R^2 \leq A_R(\|G\|_2 + b_R) + \|G\|_2 b_R.$$

The nonnegative root of this quadratic is at most $\|G\|_2 + 2b_R$. Since $b_R \rightarrow 0$ and $\eta_R \rightarrow 1$ pointwise, Fatou's lemma gives $\nabla^q \phi \in L^2(\mu)$. $\square$

B.4 A Graph Core for the Raw Adjoint

The proposition below approximates a field by smooth potential fields while also approximating its adjoint image. This lets the Hodge estimate pass from smooth fields to the full adjoint domain.

We interpret the weighted divergence in Equation (3.3) distributionally when necessary. On $\mathcal{K}_{r+1}$ its value is smooth and in $L^2(\mu)$, and integration by parts gives $\mathbb{E} \text{div}_V F = 0$.

Proposition B.4 (Raw Adjoint Core). *The operator $S_r : \mathcal{K}_{r+1} \subset \mathcal{C}_{r+1} \rightarrow \mathcal{G}_r$, defined by*

$$S_r F = P_{\mathcal{G}_r} \text{div}_V F, \quad \text{Dom}(S_r) = \mathcal{K}_{r+1} \subset \mathcal{C}_{r+1},$$

is closable and

$$\overline{S_r} = \mathcal{D}_r^*. \tag{B.4}$$

Thus, the uncentered potential core is a graph core for the raw adjoint.

Proof. Integration by parts gives $S_r \subset \mathcal{D}_r^*$, so S_r is closable. Its domain is dense by Lemma B.2. We prove $S_r^* = \mathcal{D}_r$; taking adjoints then proves Equation (B.4).

The inclusion $\mathcal{D}_r \subset S_r^*$ follows by integration by parts. For the reverse inclusion, let $u \in \text{Dom}(S_r^*) \subset \mathcal{G}_r$ and write $g = S_r^* u \in \mathcal{C}_{r+1}$. If $r \geq 1$, iterate Lemma B.1 to write $u = \nabla^r \phi$ with $\phi \in W^{r,2}(\mu)$; if $r = 0$, simply put $\phi = u$. The adjoint identity against $F = \nabla^{r+1} \psi$, $\psi \in C_c^\infty(\mathbb{R}^n)$, is

$$\int \langle \nabla^{r+1} \phi, \nabla^{r+1} \psi \rangle d\mu = \int \langle g, \nabla^{r+1} \psi \rangle d\mu.$$

Lemma B.3 shows that $\nabla u \in L^2(\mu)$. Thus, $u \in \text{Dom}(\mathcal{D}_r)$. The original adjoint identity, together with density of $\mathcal{K}_{r+1}$, now implies $\mathcal{D}_r u = g$. This proves the reverse inclusion. $\square$

B.5 The Constrained Hodge Complex and Its Projection Identity

Here, we supply the operator details behind the projection estimate. We first justify differentiation on the constrained field spaces. We then show how an inverse Laplacian measures the norm lost when projecting onto compatible fields.

Fix an integer $m \geq 1$ and use $b, \mathcal{E}_j, P_j$, and d_j from Appendix A. The spaces $\mathcal{E}_j$ are finite-dimensional subspaces of $\text{Sym}^{m-1} \mathbb{R}^n \otimes \Lambda^j \mathbb{R}^n$, independent of x . The identity $bd + db = 0$ ensures that exterior differentiation preserves the constraint defining them.

The lemma below justifies differentiation and integration by parts for these constrained fields. It also lets us approximate a field and the two derivatives appearing in the Hodge estimate at the same time.

Lemma B.5 (Closed Compatibility Complex). *Take $d_j : \text{Dom}(d_j) \subset L^2(\mu; \mathcal{E}_j) \rightarrow L^2(\mu; \mathcal{E}_{j+1})$ on its maximal distributional domain. These are closed densely defined operators satisfying $d_{j+1}d_j = 0$. Their adjoints are the maximal distributional expressions*

$$d_j^* = P_j d^*, \quad d^* = \sum_i \iota_i (-\partial_i + V_i),$$

where d_j^* maps its domain in $L^2(\mu; \mathcal{E}_{j+1})$ to $L^2(\mu; \mathcal{E}_j)$, and ι_i is exterior contraction. Smooth compactly supported sections of $\mathcal{E}_2$ are a joint graph core for d_2 and d_1^* .

Proof. Closedness follows by taking distributional limits. Smooth compactly supported sections are in the domains and are dense in L^2 . For $\omega \in \text{Dom}(d_j)$, its next distributional derivative is zero, so it lies in the next domain and $d_{j+1}d_j\omega = 0$.

For the remaining assertions, we approximate a field and all its stated operator images at the same time. First multiply the field by the cutoffs χ_R used above. The commutators with each first-order expression have pointwise norm at most $C|\nabla\chi_R||\omega|$. They tend to zero in L^2 , while the other terms converge by dominated convergence.

For a fixed compact support, convolve with a nonnegative smooth approximate identity φ_ε supported in the ball of radius ε . The coefficients of the derivative terms and the projections onto the tensor spaces are constant, so they commute with convolution. The remaining coefficients are derivatives of the potential, smooth and bounded on a fixed enlarged compact set. For each $V_i = \partial_i V$, the commutator is

$$\begin{aligned} & V_i(x)(\omega * \varphi_\varepsilon)(x) - ((V_i\omega) * \varphi_\varepsilon)(x) \\ &= \int \varphi_\varepsilon(y)[V_i(x) - V_i(x-y)]\omega(x-y) dy. \end{aligned}$$

The bracket is bounded by $C|y|$. Young's convolution inequality bounds the unweighted L^2 norm by $C\varepsilon\|\omega\|_{L^2(dx)}$. Equivalence of the weighted and unweighted norms on the enlarged compact set gives convergence in $L^2(\mu)$ as well. Convolution preserves the fixed fiber $\mathcal{E}_j$. This proves graph density, jointly for any of the stated first-order expressions.

Integration by parts on smooth compact sections identifies the formal adjoint as $P_j d^*$. Testing against compact sections first shows that every vector in the Hilbert adjoint domain has this distributional expression in L^2 . Conversely the graph approximation just proved extends the integration by parts identity to the maximal domains, giving the reverse inclusion. This proves the adjoint assertion and the joint graph-core claim. $\square$

In particular, the nonnegative quadratic form on $L^2(\mu; \mathcal{E}_2)$,

$$\ell(\omega) = \|d_1^* \omega\|_2^2 + \|d_2 \omega\|_2^2, \quad \text{Dom}(\ell) = \text{Dom}(d_1^*) \cap \text{Dom}(d_2),$$

is closed. The joint graph core extends Equation (A.7) to its entire form domain. Applying the inequality to differences of approximating compact sections makes their ordinary derivatives Cauchy. The curvature integrals converge because $Q_m(K)$ is bounded. In particular,

$$\ell(\omega) \geq \mathbb{E} \langle \omega, Q_m(K) \omega \rangle \geq \frac{m+1}{m} a \|\omega\|_2^2.$$

We can now compute the error made when projecting onto a kernel. Applied to the curl operator, the next lemma measures the loss from requiring a field to be compatible.

Lemma B.6 (Hilbert-Complex Projection Identity). *Let $\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$ be Hilbert spaces, and let $A : \text{Dom}(A) \subset \mathcal{X}_1 \rightarrow \mathcal{X}_2$ and $B : \text{Dom}(B) \subset \mathcal{X}_2 \rightarrow \mathcal{X}_3$ be closed densely defined operators with $BA = 0$, including the domain inclusion. Let $\mathcal{L} : \text{Dom}(\mathcal{L}) \subset \mathcal{X}_2 \rightarrow \mathcal{X}_2$ be the nonnegative self-adjoint operator associated with the densely defined closed form*

$$\ell(z) = \|A^* z\|^2 + \|Bz\|^2, \quad \text{Dom}(\ell) = \text{Dom}(A^*) \cap \text{Dom}(B).$$

Assume $\mathcal{L} \succeq c\text{Id}$ for some $c > 0$. For every $u \in \text{Dom}(A)$,

$$\|(\text{Id} - P_{\ker A})u\|^2 = \langle Au, \mathcal{L}^{-1} Au \rangle. \quad (\text{B.5})$$

If additionally $\ell(z) \geq \langle z, Qz \rangle$ on the form domain, where $Q : \mathcal{X}_2 \rightarrow \mathcal{X}_2$ is a bounded positive operator with $Q \succeq c\text{Id}$, then

$$\langle Au, \mathcal{L}^{-1} Au \rangle \leq \langle Au, Q^{-1} Au \rangle. \quad (\text{B.6})$$

Proof. The decomposition $\mathcal{X}_2 = \ker B \oplus (\ker B)^\perp$ splits the form ℓ into independent terms on the two summands. Indeed, $\ker B$ lies in the domain of B , and $(\ker B)^\perp \subset \ker A^*$ because $\text{ran } A \subset \ker B$. Both orthogonal projections therefore preserve both form domains. The A^* term depends only on the first component and the B term only on the second.

Put $t = Au \in \mathcal{X}_2$ and $z = \mathcal{L}^{-1} t \in \text{Dom}(\mathcal{L})$. Write $z_\perp = P_{(\ker B)^\perp} z$. This vector lies in the form domain by the preceding decomposition. Testing the weak equation for z against $z_\perp$, the two form components are orthogonal and $t \in \ker B$, so

$$c\|z_\perp\|^2 \leq \ell(z_\perp) = \langle t, z_\perp \rangle = 0.$$

Thus, $z \in \ker B$. On that subspace the form is $\|A^* z\|^2$, hence $z \in \text{Dom}(AA^*)$ and $AA^* z = t$. To verify the domain statement directly, test the equation $\mathcal{L} z = t$ against form-domain vectors in $\ker B$. Every $v \in \text{Dom}(A^*)$ has its projection onto $\ker B$ in this test class. Its complementary projection lies in $\ker A^*$ and is orthogonal to t . Thus, the same equation holds against every $v \in \text{Dom}(A^*)$, which is the defining condition $A^* z \in \text{Dom}(A)$ with $AA^* z = t$.

Now $v = A^* z \in \text{Dom}(A) \subset \mathcal{X}_1$ is orthogonal to $\ker A$, while $A(u - v) = 0$. Hence, $v = (\text{Id} - P_{\ker A})u$, and

$$\|v\|^2 = \langle z, Av \rangle = \langle \mathcal{L}^{-1} t, t \rangle,$$

which proves Equation (B.5). Finally the elementary variational formula for a positive operator gives

$$\langle t, \mathcal{L}^{-1} t \rangle = \sup_{z \in \text{Dom}(\ell)} \{2 \text{Re} \langle t, z \rangle - \ell(z)\} \leq \sup_{z \in \mathcal{X}_2} \{2 \text{Re} \langle t, z \rangle - \langle z, Qz \rangle\} = \langle t, Q^{-1} t \rangle.$$

The first identity follows by completing the square in the form norm around $\mathcal{L}^{-1}t$, and the last by completing the square around $Q^{-1}t$. $\square$

Taking $A = d_1$, $B = d_2$, and $Q = Q_m(K)$ gives the projection identity and inverse comparison used in Appendix A. Under $\mathcal{I}_m$, the kernel of d_1 is $\mathcal{C}_m$, and Equation (3.4) identifies the two projections of the weighted divergence.

C Proof of the Uniform Power Bound

We prove Lemma 5.1 from Section 5. The proof first controls the spectral radius, then obtains a single prefactor for all powers from a recurrence along adjoint orbits.

We use the following spectral form of Jensen's inequality. If $X : \mathcal{X} \rightarrow \mathcal{X}$ is a bounded positive operator and $v \in \mathcal{X}$ is a unit vector, then

$$\langle v, g(X)v \rangle \leq g(\langle v, Xv \rangle). \quad (\text{C.1})$$

Indeed, the spectral measure of X at v is a probability measure, and g is concave.

Proof of Lemma 5.1. Write $r(R) \in [0, \infty)$ for the spectral radius of R and $\sigma(R^*) \subset \mathbb{C}$ for the spectrum of its adjoint, after complexification if necessary. We first prove

$$r(R)^2 \leq \left(\frac{\gamma_D^2}{a} \right)^{1/(D+1)}. \quad (\text{C.2})$$

Complexification preserves the norms and the operator inequality, so it suffices to work over complex Hilbert spaces. If $r(R) = 0$, Equation (C.2) holds immediately. Otherwise choose $\lambda \in \sigma(R^*)$ with $|\lambda| = r(R) > 0$. We will use unit vectors $v_h \in \mathcal{X}$, indexed by positive integers h , satisfying

$$\|(R^* - \lambda \text{Id})v_h\| \longrightarrow 0. \quad (\text{C.3})$$

Put $z_h = (1 + h^{-1})\lambda \in \mathbb{C}$. These points lie outside the spectral-radius circle and hence in the resolvent set. If $|z_h - \lambda| \|(R^* - z_h \text{Id})^{-1}\| < 1$, the Neumann series would make $R^* - \lambda \text{Id}$ invertible. Therefore,

$$\|(R^* - z_h \text{Id})^{-1}\| \geq |z_h - \lambda|^{-1} \longrightarrow \infty.$$

Choose a unit vector $w_h \in \mathcal{X}$ whose resolvent image has norm at least half the resolvent norm, and normalize that image to obtain v_h . Then

$$\|(R^* - \lambda \text{Id})v_h\| \leq \frac{2}{\|(R^* - z_h \text{Id})^{-1}\|} + |z_h - \lambda| \longrightarrow 0,$$

as required.

Set $s = |\lambda|^2$. The approximate eigenvector relation implies

$$\|R^*v_h\|^2 \longrightarrow s, \quad \liminf_h \|Rv_h\|^2 \geq s.$$

The second inequality follows from $\|Rv_h\| \geq |\langle v_h, Rv_h \rangle| = |\langle R^*v_h, v_h \rangle|$. Pass to a subsequence on which $\|L^*v_h\|^2$ converges to some $p \geq 0$. Applying Equation (C.1) to $RR^* + LL^*$ gives

$$\|Rv_h\|^2 \leq \langle v_h, g(RR^* + LL^*)v_h \rangle \leq g(\|R^*v_h\|^2 + \|L^*v_h\|^2).$$

Taking the lower limit on the left and the limit on the right yields $s \leq g(s + p) < a^{-1}$, and rearranging gives

$$p \geq \frac{as^2}{1 - as}. \quad (\text{C.4})$$

For the fixed power $D - 1$, the identity

$$(R^*)^{D-1} - \lambda^{D-1} \text{Id} = \sum_{\ell=0}^{D-2} \lambda^{D-2-\ell} (R^*)^\ell (R^* - \lambda \text{Id})$$

and Equation (C.3) show that

$$L^*(R^*)^{D-1} v_h - \lambda^{D-1} L^* v_h \longrightarrow 0.$$

Since $\|L^*(R^*)^{D-1}\| = \|R^{D-1}L\| \leq \gamma_D$, this implies $s^{D-1}p \leq \gamma_D^2$. Combining it with Equation (C.4) gives

$$\frac{as^{D+1}}{1 - as} \leq \gamma_D^2.$$

In particular $as^{D+1} \leq \gamma_D^2$, proving Equation (C.2).

The spectral-radius bound controls asymptotic growth. We next obtain one prefactor valid for every iterate. Fix B as in the statement and a unit vector $f \in \mathcal{X}$. For integers $m \geq 0$, define the nonnegative real sequences

$$x_m = \|(R^*)^m f\|^2, \quad b_m = \|L^*(R^*)^m f\|^2, \quad y_m = \frac{x_m}{B^m} \quad (m \geq 0).$$

For $m \geq 1$ with $x_{m-1}, x_m > 0$, Cauchy–Schwarz gives

$$x_m^2 = \left| \left\langle (R^*)^{m-1} f, R(R^*)^m f \right\rangle \right|^2 \leq x_{m-1} \|R(R^*)^m f\|^2.$$

Apply Equations (5.1) and (C.1) to the unit vector $(R^*)^m f / \sqrt{x_m}$. After division by x_m , the last inequality becomes

$$\frac{x_m}{x_{m-1}} \leq g\left(\frac{x_{m+1} + b_m}{x_m}\right).$$

The argument of g is finite, so $ax_m/x_{m-1} < 1$. For $0 \leq v < a^{-1}$,

$$g^{-1}(v) = \frac{v}{1 - av} = v + \frac{av^2}{1 - av} \geq v + av^2.$$

Inverting the preceding inequality and multiplying by x_m gives

$$x_{m+1} \geq \frac{x_m^2}{x_{m-1}} + a \frac{x_m^3}{x_{m-1}^2} - b_m.$$

Equivalently,

$$y_{m+1} \geq \frac{y_m^2}{y_{m-1}} + aB \frac{y_m^3}{y_{m-1}^2} - \frac{b_m}{B^{m+1}}. \quad (\text{C.5})$$

The middle term is the gain supplied by $a > 0$. The observation bounds control the last term.

The spectral bound ensures that the normalized squared norms attain a maximum. Indeed, by Equation (C.2) and the strict inequality on B , $r(R) < \sqrt{B}$. Choose $\rho \in \mathbb{R}$ with $r(R) < \rho < \sqrt{B}$.

The spectral-radius formula gives $\|R^m\| \leq \rho^m$ for all sufficiently large m , so $y_m \leq (\rho^2/B)^m \rightarrow 0$. Since $y_0 = 1$, the sequence (y_m) attains a positive global maximum $Y_* \in [1, \infty)$ at a finite index $h \in \mathbb{N}_0$.

We first rule out $h \geq D - 1$. The last observation bound gives

$$\begin{aligned} b_h &= \left\| L^*(R^*)^{D-1} (R^*)^{h-D+1} f \right\|^2 \\ &\leq \gamma_D^2 x_{h-D+1} \leq \gamma_D^2 B^{h-D+1} Y_*. \end{aligned}$$

Every iterate preceding $x_h > 0$ is positive: a zero iterate would make every subsequent one zero. Thus, Equation (C.5) applies at h . Using $y_{h-1} \leq Y_* = y_h$ and the last bound, we obtain

$$y_{h+1} \geq Y_* + \left(aB - \frac{\gamma_D^2}{B^D} \right) Y_* > Y_*,$$

contrary to maximality. Hence, every global maximum occurs among the indices $0, \dots, D - 2$.

It remains to bound a maximum in this finite initial interval. If $h = 0$, then $Y_* = 1 \leq Q_D(B)$. Otherwise define the real differences

$$\Delta_j = y_j - y_{j-1}.$$

For $1 \leq j \leq h$, the observation bounds give $b_j \leq \gamma_{j+1}^2$. Dropping the nonnegative curvature term from Equation (C.5) and using

$$\frac{v^2}{u} - 2v + u = \frac{(v - u)^2}{u} \geq 0 \quad (u > 0)$$

therefore yields

$$\Delta_{j+1} \geq \Delta_j - \frac{\gamma_{j+1}^2}{B^{j+1}}.$$

Since $\Delta_{h+1} \leq 0$, summing this inequality from j to h gives $\Delta_j \leq \sum_{i=j}^h \gamma_{i+1}^2 / B^{i+1}$. Summing once more gives

$$Y_* - 1 = \sum_{j=1}^h \Delta_j \leq \sum_{j=1}^h \sum_{i=j}^h \frac{\gamma_{i+1}^2}{B^{i+1}} = \sum_{i=1}^h \frac{i \gamma_{i+1}^2}{B^{i+1}} \leq Q_D(B) - 1.$$

Only denominators belonging to positive iterates up to h have been used, so this argument also covers $x_{h+1} = 0$. We have proved $y_k \leq Q_D(B)$ for every k . Taking the supremum over unit f , and using $\|(R^*)^k\| = \|R^k\|$, proves Equation (5.2). $\square$

D Localization: Existence and Variance

In this appendix, we justify global existence of the localization process and the moving Appell variance estimate. We use the notation of Section 6.

D.1 Global Existence

Proof of Lemma 6.1. Let $\zeta \in (0, \infty]$ be the stopping time equal to the maximal lifetime of the local solution of Equation (6.1), and fix a deterministic time $T > 0$. We show that the covariance

stays invertible and the parameters remain finite up to this time, so the process can continue. Boundedness will then give the martingale assertion for test functions.

For $t < \zeta$, matrix Itô's formula and cyclicity of the trace give

$$\begin{aligned} d \log \det A_t &= \text{Tr}(A_t^{-1} dA_t) - \frac{1}{2} \sum_i \text{Tr}(A_t^{-1} U_{i,t} A_t^{-1} U_{i,t}) dt \\ &= \sum_i \text{Tr}(S_{i,t}) dB_{i,t} - \left(n + \frac{1}{2} \sum_i \text{Tr}(S_{i,t}^2) \right) dt. \end{aligned}$$

Here, $\sum_i \text{Tr}(S_{i,t}^2) \leq 8n$, while

$$\sum_i (\text{Tr } S_{i,t})^2 \leq n \sum_i \|S_{i,t}\|_{\text{HS}}^2 \leq 8n^2.$$

The martingale in the log-determinant equation thus has quadratic variation at most $8n^2 t$. Extend its integrand by zero after ζ . This defines a square-integrable martingale on every finite interval, with continuous paths. The drift integral also has a finite limit at any finite lifetime, since its rate lies between $-5n$ and $-n$. Consequently, on the event $\zeta \leq T$, the function $\log \det A_t$ has a finite limit as $t \uparrow \zeta$ and is bounded below for $0 \leq t < \zeta$.

Every eigenvalue of A_t is at most R^2 , because all posterior laws are supported in the same ball. The lower bound on the determinant therefore implies

$$\lambda_{\min}(A_t) \geq \frac{\det A_t}{R^{2(n-1)}} > 0$$

uniformly for $t < \zeta$, with a random positive lower bound on the event $\zeta \leq T$. Thus, A_t^{-1} is bounded along each such path.

Since $|m_t| \leq R$, both drift coefficients in Equation (6.1) are bounded along the path. The quadratic variation of the martingale part of θ_t also remains finite. To justify convergence, stop this martingale when $\|A_t^{-1}\|_{\text{op}}$ first exceeds an integer N . Extend the stopped integrand by zero after ζ . The stopped martingale has quadratic variation at most nNT on $[0, T]$. Thus, it has a continuous extension to that interval. On each path under consideration some N is never reached before ζ . The unstopped martingale consequently has a finite limit there.

The parameters θ_t and Λ_t therefore have finite limits at ζ . At those finite parameters the covariance is positive definite and the coefficients in Equation (6.1) are smooth. Local existence starting from that limiting state extends the solution past ζ , contradicting maximality. Since T was arbitrary, $\zeta = \infty$ almost surely.

For each bounded Borel function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the local martingale $\mathbb{E}_t f$ is bounded by $\|f\|_\infty$. Apply its martingale identity along a localizing sequence and pass to the limit by bounded convergence. The same passage to the limit holds after multiplication by any bounded $\mathcal{F}_s$ -measurable test. This gives $\mathbb{E}[\mathbb{E}_t f \mid \mathcal{F}_s] = \mathbb{E}_s f$ for $s \leq t$. Thus, Equation (6.3) is a true martingale identity, as required. $\square$

D.2 Lowering Operators and the Drift

Recall that the polynomial $p_t = P_d^{\mu_t}[T]$ has coefficients that change with μ_t . We compute how these changes and the changing density affect its variance. Combining the drift terms before taking norms leaves an error controlled by lower-degree polynomials, as needed for Lemma 6.2.

The calculation has three parts. We differentiate the Appell generating function to find how the polynomial changes. Multiplying by the evolving density adds an Itô cross-variation term. Identities for operators that raise and lower polynomial degree separate the first and last terms of the resulting sum from the lower-degree terms. Writing the real-valued drift process of the polynomial variance as D_t , we obtain

$$D_t \geq -2(d+1)v_t - 2d!\Sigma_d\sqrt{v_t}\|(A_t^{1/2})^{\otimes d}T\|.$$

The next subsection uses compact support and Bessel's inequality to justify taking expectations in this estimate.

At a fixed time, suppress the subscript t . Let $\Psi(z) = \log \mathbb{E}_\mu e^{z \cdot X} \in \mathbb{R}[[z_1, \dots, z_n]]$ be the formal logarithm of the moment series, and put $\Gamma_j = D^j \Psi(0) \in \text{Sym}^j \mathbb{R}^n$ for $j \geq 1$. These are the cumulant tensors (Anshelevich, 2004, Section 2.1.3). Define linear operators $\mathcal{C}_i, \mathcal{R}_i : \mathbb{R}[x_1, \dots, x_n] \rightarrow \mathbb{R}[x_1, \dots, x_n]$ that lower and raise polynomial degree:

$$\mathcal{C}_i = \sum_{k \geq 1} \frac{1}{k!} \left\langle A^{-1/2} e_i, \Gamma_{k+1}[D^k] \right\rangle, \quad \mathcal{R}_i = \xi_i - \mathcal{C}_i. \quad (\text{D.1})$$

Contracting k indices identifies Γ_{k+1} with a map $\text{Sym}^k \mathbb{R}^n \rightarrow \mathbb{R}^n$. Thus, $\Gamma_{k+1}[D^k]$ is a vector-valued differential operator, paired with $A^{-1/2} e_i \in \mathbb{R}^n$. Every term in the sum lowers degree, so only finitely many terms act on a given polynomial. The term ξ_i in $\mathcal{R}_i$ denotes multiplication by that coordinate. The symbol of $\mathcal{C}_i$, obtained by replacing derivatives by the corresponding variables, is the formal series $u_i(z) = [A^{-1/2}(\nabla \Psi(z) - m)]_i \in \mathbb{R}[[z_1, \dots, z_n]]$.

By Equation (6.3), the moment series $\mathcal{Z}_t(z) = \mathbb{E}_t e^{z \cdot X} \in \mathbb{R}[[z_1, \dots, z_n]]$ satisfies $d\mathcal{Z}_t(z) = \mathcal{Z}_t(z)u_t(z) \cdot dB_t$. Itô's rule for its inverse gives

$$d\left(\frac{e^{z \cdot x}}{\mathcal{Z}_t(z)}\right) = -\frac{e^{z \cdot x}}{\mathcal{Z}_t(z)}u_t(z) \cdot dB_t + \frac{e^{z \cdot x}}{\mathcal{Z}_t(z)}|u_t(z)|^2 dt.$$

Here, $u_t = (u_{1,t}, \dots, u_{n,t})$ is a vector of formal series. Comparing degree- d coefficients in Equation (2.2) yields

$$dp_t = -\sum_i \mathcal{C}_i p_t dB_{i,t} + \sum_i \mathcal{C}_i^2 p_t dt. \quad (\text{D.2})$$

At each fixed time, a constant-coefficient differential operator with symbol $u_i(z)$ sends $e^{z \cdot x}/\mathcal{Z}_t(z)$ to $u_i(z)e^{z \cdot x}/\mathcal{Z}_t(z)$, and its square multiplies by $u_i(z)^2$. Thus, the inverse-series Itô formula gives Equation (D.2) by comparing finite Taylor coefficients.

For the product with the density, Itô's rule first gives

$$d(p_t^2) = -2 \sum_i p_t \mathcal{C}_i p_t dB_{i,t} + \left\{ \sum_i (\mathcal{C}_i p_t)^2 + 2p_t \sum_i \mathcal{C}_i^2 p_t \right\} dt.$$

The cross variation with Equation (6.2) contributes $-2 \sum_i p_t \xi_i \mathcal{C}_i p_t \mu_t(dx) dt$. Integrating these three drift terms gives the drift D_t of v_t :

$$\begin{aligned} D_t &= \sum_i \mathbb{E}_t(\mathcal{C}_i p_t)^2 + 2 \sum_i \mathbb{E}_t[p_t \mathcal{C}_i^2 p_t] - 2 \sum_i \mathbb{E}_t[p_t \xi_i \mathcal{C}_i p_t] \\ &= \sum_i \|\mathcal{C}_i p_t\|_{L^2(\mu_t)}^2 - 2 \left\langle p_t, \sum_i \mathcal{R}_i \mathcal{C}_i p_t \right\rangle_{L^2(\mu_t)}. \end{aligned} \quad (\text{D.3})$$

Differentiating the Appell generating function with respect to z in the direction $A^{-1/2}e_i$ gives

$$D_{A^{-1/2}e_i}^z \left(\frac{e^{z \cdot x}}{\mathcal{Z}(z)} \right) = (\xi_i - u_i(z)) \frac{e^{z \cdot x}}{\mathcal{Z}(z)}.$$

For $S \in \text{Sym}^j \mathbb{R}^n$, coefficient comparison therefore identifies the raising operator:

$$\mathcal{R}_i P_j[S] = P_{j+1}[\text{Sym}(A^{-1/2}e_i \otimes S)]. \quad (\text{D.4})$$

To estimate the lowering operator, we express the cumulants through Appell polynomials. The following identity holds for any covariance. Let $k \geq 1$ be an integer, let $U \in \text{Sym}^k \mathbb{R}^n$, and let μ be any probability law on $\mathbb{R}^n$ with moments through order $k+1$. For X with law μ , write $m = \mathbb{E}_\mu X \in \mathbb{R}^n$. Here, Ψ is the formal logarithm of the moment series and Γ_j is its degree- j cumulant tensor. Comparing degree- k coefficients in the formal identity

$$\frac{\mathbb{E}_\mu[X e^{z \cdot X}]}{\mathbb{E}_\mu e^{z \cdot X}} = \nabla \Psi(z)$$

and using Equation (2.2) gives

$$\Gamma_{k+1}[U] = \mathbb{E}_\mu[X P_k^\mu[U]] = \mathbb{E}_\mu[(X - m) P_k^\mu[U]].$$

The last equality uses $\mathbb{E}_\mu P_k^\mu[U] = 0$ for $k \geq 1$. Only finite Taylor coefficients are involved; convergence of the moment-generating function is not required.

For an isotropic law with moments through order $2k$, the coordinate functions are orthonormal in $L^2(\mu)$. Bessel's inequality therefore gives

$$|\Gamma_{k+1}[U]|^2 = \sum_i (\mathbb{E}_\mu[X_i P_k^\mu[U]])^2 \leq \mathbb{E}_\mu P_k^\mu[U]^2 \leq (k! c_k(\mu))^2 \|U\|^2,$$

and hence

$$\|\Gamma_{k+1} : \text{Sym}^k \mathbb{R}^n \longrightarrow \mathbb{R}^n\|_{\text{op}} \leq k! c_k(\mu). \quad (\text{D.5})$$

We apply this estimate after centering the posterior and rescaling it to have identity covariance, a change of variables called whitening. At the fixed time under consideration, write $y = A^{-1/2}(x - m)$, $\bar{\mu} = \mathcal{L}(\xi)$, and $\bar{p}(y) = p_t(m + A^{1/2}y)$. Equation (2.6) gives $\bar{p} = P_d^\mu[\tilde{T}]$, where the leading tensor is $\tilde{T} = (A^{1/2})^{\otimes d} T \in \text{Sym}^d \mathbb{R}^n$.

We also track the operators under this change of variables. The cumulant generator of $\bar{\mu}$ is $\Psi_\xi(z) = \Psi(A^{-1/2}z) - z \cdot A^{-1/2}m$, so its cumulants satisfy

$$\Gamma_j^\xi = (A^{-1/2})^{\otimes j} \Gamma_j \quad (j \geq 2).$$

The chain rule gives $D_x = A^{-1/2}D_y$. Thus, the k derivative slots in each term of $\mathcal{C}_i$, together with its first contraction against $A^{-1/2}e_i$, transform all $k+1$ slots of the cumulant. The corresponding operators on polynomials in y are

$$\bar{\mathcal{C}}_i = \sum_{k \geq 1} \frac{1}{k!} \langle e_i, \Gamma_{k+1}^\xi[D_y^k] \rangle, \quad \bar{\mathcal{R}}_i = y_i - \bar{\mathcal{C}}_i.$$

In particular, $(\mathcal{C}_i p_t)(m + A^{1/2}y) = \bar{\mathcal{C}}_i \bar{p}(y)$, and the same identity holds for the raising operators. These are identities at a fixed time, used to estimate the drift already computed in the original

coordinates. The whitened law is isotropic and log-concave, so its degree- k coefficient is at most C_k for $1 \leq k < d$ by the hypotheses of Lemma 6.2.

We now decompose $\sum_i \mathcal{R}_i \mathcal{C}_i p_t$ according to the index k in Equation (D.1). Since $\Gamma_2^\xi = \text{Id}$, the $k = 1$ part of $\mathcal{C}_i$ is ∂_{y_i} . The derivative and raising identities give

$$\begin{aligned} \sum_i \overline{\mathcal{R}}_i \partial_{y_i} \bar{p} &= d P_d^{\bar{\mu}} \left[\sum_i \text{Sym}(e_i \otimes \tilde{T}(e_i, \cdot, \dots, \cdot)) \right] \\ &= d \bar{p}. \end{aligned}$$

The sum inside the brackets reconstructs $\tilde{T}$ by expanding one slot in the coordinate basis. Thus, this contribution is $d p_t$ in the original coordinates.

For $k = d$, the identity $D_y^d \bar{p} = d! \tilde{T}$ cancels the factor $1/d!$ in the lowering operator. The resulting vector of constant polynomials is $\Gamma_{d+1}^\xi[\tilde{T}] = \mathbb{E}_t[\xi_t p_t]$, by the cumulant identity above. Applying a raising operator to a constant multiplies it by the corresponding coordinate. Hence, the $k = d$ contribution is $\langle \mathbb{E}_t[p_t \xi_t], \xi_t \rangle$. Its pairing with p_t is $|\mathbb{E}_t[p_t \xi_t]|^2 \leq v_t$ by Bessel's inequality. Estimating this endpoint by v_t leaves only coefficients of degree below d on the right, as required for induction.

For $2 \leq k \leq d-1$, put $j = d - k + 1$ and work in the whitened variables. Contracting k slots of $\tilde{T}$ into Γ_{k+1}^ξ defines $W_k \in \mathbb{R}^n \otimes \text{Sym}^{d-k} \mathbb{R}^n$ by

$$(W_k)_{iJ} = \sum_I (\Gamma_{k+1}^\xi)_{iI} \tilde{T}_{IJ}, \quad |I| = k, \quad |J| = d - k.$$

For each ordered J , the slice $\tilde{T}_{\cdot J}$ is an element of $\text{Sym}^k \mathbb{R}^n$. Thus, Equation (D.5) gives

$$\|W_k\|^2 = \sum_J |\Gamma_{k+1}^\xi[\tilde{T}_{\cdot J}]|^2 \leq (k! C_k)^2 \sum_J \|\tilde{T}_{\cdot J}\|^2 = (k! C_k)^2 \|\tilde{T}\|^2.$$

Taking k derivatives and dividing by $k!$ contributes $d!/[k!(d-k)!]$. The raising identity in Equation (D.4) then turns the k th contribution to $\sum_i \mathcal{R}_i \mathcal{C}_i p_t$, viewed as a polynomial of ξ , into

$$\frac{d!}{k!(d-k)!} P_j^{\mathcal{L}(\xi)} [\text{Sym } W_k].$$

Orthogonal symmetrization has norm at most one. Applying the degree- j coefficient bound therefore gives the L^2 estimate

$$\frac{d!}{k!(d-k)!} (k! C_k) (j! C_j) \|\tilde{T}\| = d! j C_k C_j \|\tilde{T}\|.$$

Dropping the nonnegative first term of Equation (D.3) now gives

$$D_t \geq -2(d+1)v_t - 2d! \Sigma_d \sqrt{v_t} \|\tilde{T}_t\|. \quad (\text{D.6})$$

D.3 Integrability and Passage to Expectations

Compact support lets us control the stochastic terms well enough to take expectations. We then integrate the resulting variance inequality to finish the proof of Lemma 6.2.

The localization process is global by Lemma 6.1. Every posterior remains in B_R . For each monomial f of degree at most $2d$, Equation (6.3) expresses its moment as a martingale with diffusion coefficient $\mathbb{E}_t[f\xi_t]$. Since the coordinates of ξ_t are orthonormal, Bessel's inequality gives

$$|\mathbb{E}_t[f\xi_t]|^2 \leq \mathbb{E}_t f^2 \leq \sup_{x \in B_R} |f(x)|^2.$$

Thus, the finite vector of moments through order $2d$ and all its diffusion coefficients are uniformly bounded by deterministic constants.

By Equation (2.2), $v_t = \mathbb{E}_t(P_d^{\mu_t}[T])^2$ is a fixed polynomial in that moment vector. Its first two derivatives are bounded on a compact box containing those moment values. Itô's formula therefore gives deterministic bounds for both the martingale coefficient and the drift of v_t , depending only on R, n, d, T . Uniqueness of the semimartingale decomposition identifies this drift with D_t in Equation (D.3).

First perform these calculations with the tilt parameters stopped before they leave bounded sets. Global existence makes these stopping times increase to infinity. The stopped stochastic integrals converge in L^2 , and the drift integrals converge in L^1 , by these deterministic bounds. The bounded variables v_t also converge in L^1 . Consequently $\mathcal{V}(t) = \mathbb{E}v_t$ is locally absolutely continuous, with derivative $\mathbb{E}D_t$ almost everywhere. Taking expectations in Equation (D.6) and applying Cauchy–Schwarz gives

$$\mathcal{V}'(t) \geq -2(d+1)\mathcal{V}(t) - 2d!\Sigma_d \sqrt{\mathcal{V}(t)} \sqrt{\mathbb{E} \|\tilde{T}_t\|^2} \quad \text{for almost every } t.$$

By Equation (6.6) with all times equal to t , the last square root is at most $e^{2d^2t} \|T\|$. If $T \neq 0$, the leading polynomial is nonzero on each full-dimensional posterior, so $\mathcal{V}(t) > 0$. Local absolute continuity makes $\mathcal{V}$ continuous, so it has a positive minimum on each compact time interval. Division by $2\sqrt{\mathcal{V}(t)}$ is therefore legitimate there. Multiplication by $e^{(d+1)t}$ and integration give

$$\begin{aligned} \sqrt{\mathcal{V}(0)} &\leq e^{(d+1)t} \sqrt{\mathcal{V}(t)} + d!\Sigma_d \|T\| \int_0^t e^{(2d^2+d+1)s} ds \\ &\leq e^{(d+1)t} \sqrt{\mathcal{V}(t)} + d!\Sigma_d t e^{(2d^2+d+1)t} \|T\|. \end{aligned}$$

This is Equation (6.9). The case $T = 0$ is immediate.

E Proof of Uniform Coefficient Growth

We prove Theorem 8.1. We use the weight w , the fixed radius R_* , and the targets β_d defined in Section 8.

The numerical choices are listed in Table 1: C_w, R_* are positive real constants and D_0 is a positive integer. The one-derivative step handles all degrees at most D_0 . Above this cutoff, the decay of $w(D)^2$ makes the final observation small enough to apply the power lemma at a rate below R_*^2 . We verify its strict hypothesis and its prefactor separately.

Proof of Theorem 8.1. The covariance bound gives the base case:

$$c_1^* \leq 1 < \frac{R_*}{16} = \beta_1.$$

Parameter	Value	Role
C_w	4	Bound for convolution of the weights
R_*	10^8	Fixed coefficient radius
D_0	10^7	Last degree handled by one derivative

Table 1: Constants for the coefficient induction.

We first record the convolution bound needed for the drift. For every integer $s \geq 1$,

$$\sum_{k=1}^s \frac{w(k)w(s-k)}{w(s)} \leq C_w. \quad (\text{E.1})$$

The endpoint $k = s$ contributes one. If $1 \leq k \leq s/2$, then $(s+1)/(s-k+1) \leq 2$, so the corresponding summand is at most $16/(k+1)^4$. Pairing terms on opposite sides of the midpoint therefore yields

$$\begin{aligned} \sum_{k=1}^s \frac{w(k)w(s-k)}{w(s)} &\leq 1 + 32 \sum_{j=2}^{\infty} j^{-4} \\ &\leq 1 + 32 \left(\frac{1}{16} + \frac{1}{81} + \int_3^{\infty} x^{-4} dx \right) = \frac{307}{81} < 4 = C_w. \end{aligned}$$

Fix $d \geq 2$ and assume the conclusion in every smaller degree. These lower-degree coefficients are therefore finite, so we may use $C_k = \beta_k$ in Proposition 7.1. Then

$$\Sigma_d = \sum_{k=2}^{d-1} (d-k+1) \beta_k \beta_{d-k+1}.$$

Every coefficient in this sum has degree less than d . Its product is $R_*^{d+1} w(k)w(d-k+1)$, and $d-k+1 \leq d$. Applying Equation (E.1) at $s = d+1$ gives

$$\Sigma_d \leq d C_w R_*^{d+1} w(d+1). \quad (\text{E.2})$$

For $d = 2$ the sum is empty, so $\Sigma_2 = 0$ and the same bound holds.

Degrees Up to the Cutoff. Suppose $2 \leq d \leq D_0$. In this range, the quadratic seed controls a single integration. We choose the localization time so that both the integration term and the drift fit within the target bound:

$$q = 1, \quad \eta = \frac{d}{100R_*}, \quad \delta = \frac{\eta^2}{d^2} = (100R_*)^{-2}.$$

$$0 < \eta \leq \frac{D_0}{100R_*} = \frac{1}{1000} < \frac{1}{100}.$$

Corollary 5.3, with $a = \delta$, supplies the admissible integration bound

$$M_1 = (2/\delta)^{1/6} = [2(100R_*)^2]^{1/6} \leq 10\sqrt{R_*}.$$

The last inequality follows by taking sixth powers: $20000R_*^2 \leq 10^6 R_*^3$. Also, $(d+1)/d \leq 3/2$ gives

$$\frac{\beta_{d-1}}{\beta_d} = \frac{1}{R_*} \left(\frac{d+1}{d} \right)^4 < \frac{16}{R_*}.$$

For the drift, Equation (E.2) gives

$$\frac{\eta \Sigma_d}{d^2 \beta_d} \leq \frac{\eta C_w R_*}{d} = \frac{1}{25}.$$

Thus, dividing Equation (7.2) by β_d yields

$$\begin{aligned} \frac{c_d^*}{\beta_d} &\leq e^{3\eta} \left[\sqrt{1+\eta} \frac{160}{\sqrt{R_*}} + \frac{1}{25} \right] \\ &< \frac{21}{20} \left[\frac{21}{20} \frac{2}{125} + \frac{1}{25} \right] = \frac{1491}{25000} < \frac{3}{50}. \end{aligned}$$

Here, $e^{3\eta} \leq (1 - 3\eta)^{-1} < 21/20$ and $\sqrt{1+\eta} < 21/20$ follow from $\eta < 1/100$. This proves the induction step through degree D_0 .

Degrees Above the Cutoff. Suppose $d > D_0$. We take about half the derivatives and use the power estimate with coefficients only through degree $\lfloor \sqrt{d} \rfloor$. The localization parameter is chosen to make the drift small:

$$q = \lfloor d/2 \rfloor, \quad D = \lfloor \sqrt{d} \rfloor, \quad \eta = \frac{d^{-1/2}}{100C_w R_*}, \quad \delta = \frac{\eta^2}{d^2} = \frac{d^{-3}}{(100C_w R_*)^2}. \quad (\text{E.3})$$

These choices satisfy $1 \leq q \leq d-1$, $2 \leq D < d$, and $0 < \eta < 1/100$; in particular, $0 < \delta < 1$. For every regular centered law ν with covariance at most Id and curvature at least δ , the induction hypothesis and Equation (4.12) give

$$\|J_\nu^{j-1} \mathbf{L}_\nu\| \leq c_j^* \leq \gamma_j := \frac{R_*^j}{(j+1)^4}, \quad 2 \leq j \leq D.$$

The operator comparison in Equation (4.11) also holds with $a = \delta$. We apply Lemma 5.1 directly, with

$$B = R_*^2 e^{-1/(D+1)}.$$

Its strict hypothesis follows from

$$\delta B^{D+1} = \frac{R_*^{2D}}{e(100C_w)^2 d^3} > \frac{R_*^{2D}}{(D+1)^8} = \gamma_D^2. \quad (\text{E.4})$$

Indeed, $D+1 > \sqrt{d}$ and $e(100C_w)^2 < 3 \cdot 400^2 = 480000 < D_0 < d$.

We also retain the prefactor in that lemma rather than bounding each integration separately. Its value satisfies

$$\begin{aligned} Q_D(B) &= 1 + \sum_{j=1}^{D-2} \frac{j \gamma_{j+1}^2}{B^{j+1}} \\ &= 1 + \sum_{j=1}^{D-2} \frac{j e^{(j+1)/(D+1)}}{(j+2)^8} \\ &\leq 1 + e \sum_{j=1}^{\infty} (j+2)^{-7} \leq 1 + e \int_2^{\infty} x^{-7} dx = 1 + \frac{e}{384} < 2. \end{aligned} \quad (\text{E.5})$$

Consequently, for every law in the preceding class,

$$\|J_\nu^q\| \leq M_q := \sqrt{2} R_*^q \exp \left[-\frac{q}{2(D+1)} \right]. \quad (\text{E.6})$$

This is the uniform input required by Proposition 7.1.

We divide each of the two terms in the transfer inequality by the target bound β_d . Substituting the chosen M_q and $\beta_j = R_*^j w(j)$ gives the exact factorization of the first ratio:

$$\begin{aligned} & \frac{e^{3\eta}(1+\eta)^{q/2}M_q\beta_{d-q}}{\beta_d} \\ &= \sqrt{2} \frac{w(d-q)}{w(d)} \exp \left[3\eta + \frac{q}{2} \log(1+\eta) - \frac{q}{2(D+1)} \right]. \end{aligned}$$

The exponential factors R_*^d have canceled. Since $d-q \geq d/2$,

$$\frac{w(d-q)}{w(d)} = \left(\frac{d+1}{d-q+1} \right)^4 \leq 16.$$

Using also $\log(1+\eta) \leq \eta$, the resulting ratio, including the factor $e^{3\eta}$, is at most

$$32 \exp \left\{ 3\eta + q \left[\frac{\eta}{2} - \frac{1}{2(D+1)} \right] \right\}. \quad (\text{E.7})$$

Here, 32 bounds $16\sqrt{2}$. We have $D+1 \leq 2\sqrt{d}$, and hence

$$\eta(D+1) \leq \frac{2}{100C_w R_*} < \frac{1}{2}.$$

Therefore, the bracket in Equation (E.7) is at most $-1/[4(D+1)]$. Using $q \geq d/3$, $D+1 \leq 2\sqrt{d}$, and $\eta < 1/100$, the principal ratio is at most

$$32 \exp \left(\frac{3}{100} - \frac{\sqrt{d}}{24} \right) < \frac{1}{4}.$$

For completeness, $d > D_0 > 10^6$ gives $\sqrt{d} > 1000$, so the exponent is less than -7 ; then $e > 2$ gives $32e^{-7} < 1/4$.

For the second term, Equation (E.2) and $w(d+1) \leq w(d)$ imply

$$\frac{e^{3\eta}\eta\Sigma_d}{d^2\beta_d} \leq e^{3\eta}\frac{\eta C_w R_*}{d} = \frac{e^{3\eta}}{100d^{3/2}} < \frac{1}{50}.$$

The two ratios sum to less than one. Hence, Equation (7.2) gives $c_d^* < \beta_d$. □

F Approximation and Extension to General Test Functions

We carry out the approximation in two steps. First, we approximate an isotropic log-concave measure by smooth isotropic measures with positive lower and finite upper curvature bounds. Every fixed moment converges. We then extend the Poincaré inequality from smooth compactly supported functions to functions whose squared gradient is integrable.

F.1 Moments and Gaussian Regularization

Approximation requires control of both moments and curvature. We first show that log-concave laws have all polynomial moments. We then check that adding a small Gaussian gives a smooth density with controlled curvature and moments converging to those of the original measure.

Let X be an $\mathbb{R}^n$ -valued random vector with a full-dimensional log-concave law μ . For each coordinate, Prékopa's marginal theorem (Prékopa, 1973, Theorem 6) shows that the survival function $F : \mathbb{R} \rightarrow [0, 1]$, $F(t) = \Pr\{X_i \geq t\}$, is log-concave: integrate the jointly log-concave function $\mathbf{1}_{\{x_i \geq t\}} d\mu/dx$ in x . If the upper tail is unbounded, choose real numbers $a < b$ with $0 < F(b) < F(a)$. Concavity of $\log F$, applied to a, b, t , gives

$$F(t) \leq F(a) \exp\left(\frac{t-a}{b-a} \log \frac{F(b)}{F(a)}\right), \quad t \geq b.$$

If the upper tail is bounded, an exponential tail bound already holds beyond its endpoint. Applying the same argument to $-X_i$ gives an exponential bound for $\Pr\{|X_i| > t\}$. Since $|X| > t$ implies $|X_i| > t/\sqrt{n}$ for some i , the union bound gives an exponential tail bound for $|X|$. The identity $\mathbb{E}|X|^k = \int_0^\infty kt^{k-1} \Pr\{|X| > t\} dt$ then proves finiteness for every $k > 0$.

The same conclusion holds on the affine support of any log-concave law. In dimension zero it is immediate.

Adding a small Gaussian makes a density smooth. The following lemma tracks its curvature and moments during this smoothing. Its normalized version also preserves the covariance and curvature bounds we need.

Lemma F.1 (Gaussian Regularization). *Let μ be a full-dimensional log-concave law on $\mathbb{R}^n$ with curvature at least $a\text{Id}$, where $a \geq 0$. Let $X \sim \mu$ and $G \sim N(0, \text{Id})$ be independent random vectors in $\mathbb{R}^n$. For every $\varepsilon > 0$, the law of $X + \sqrt{\varepsilon}G$ has density e^{-V_ε} , with $V_\varepsilon \in C^\infty(\mathbb{R}^n)$ satisfying*

$$\frac{a}{1+a\varepsilon}\text{Id} \preceq D^2V_\varepsilon \preceq \varepsilon^{-1}\text{Id}.$$

As $\varepsilon \downarrow 0$, these laws converge weakly and in every fixed moment to μ .

If μ is centered, $\text{Cov}(\mu) \preceq \text{Id}$, and $0 < a \leq 1$, then the normalized laws

$$\mu_\varepsilon = \mathcal{L}\left(\frac{X + \sqrt{\varepsilon}G}{\sqrt{1+\varepsilon}}\right) \tag{F.1}$$

are regular. They have covariance at most Id and curvature at least $a\text{Id}$. They also converge weakly and in every fixed moment to μ .

Proof. Let G be an independent standard Gaussian vector and fix $\varepsilon > 0$. The random vector $X + \sqrt{\varepsilon}G$ has the density $\rho_\varepsilon : \mathbb{R}^n \rightarrow (0, \infty)$ given by

$$\rho_\varepsilon(x) = (2\pi\varepsilon)^{-n/2} \int \exp\left(-\frac{|x-y|^2}{2\varepsilon}\right) d\mu(y).$$

Every derivative of the Gaussian kernel is bounded as a function of $x - y$; on a compact set of x the same holds uniformly. Differentiation under the integral is therefore justified by dominated convergence, to every order.

For the lower Hessian bound, write the density of μ as $e^{-a|y|^2/2-W(y)}$ up to normalization, with $W : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ convex. This includes $a = 0$. The identity

$$\frac{a}{2}|y|^2 + \frac{|x-y|^2}{2\varepsilon} = \frac{a|x|^2}{2(1+a\varepsilon)} + \frac{1+a\varepsilon}{2\varepsilon} \left| y - \frac{x}{1+a\varepsilon} \right|^2$$

extracts a Gaussian factor from the convolution. The remaining integrand is jointly log-concave in (x, y) , so its integral is log-concave by Prékopa's marginal theorem (Prékopa, 1973, Theorem 6). Hence, $V_\varepsilon = -\log \rho_\varepsilon$ is smooth and satisfies $D^2V_\varepsilon \succeq a(1+a\varepsilon)^{-1}\text{Id}$.

The upper Hessian bound follows from an explicit calculation. At each $x \in \mathbb{R}^n$, define the probability measure $\pi_{\varepsilon,x}$ on $\mathbb{R}^n$ by

$$d\pi_{\varepsilon,x}(y) = \frac{\exp(-|x-y|^2/(2\varepsilon)) d\mu(y)}{\int \exp(-|x-z|^2/(2\varepsilon)) d\mu(z)}.$$

Write $b_i(x) = \int y_i d\pi_{\varepsilon,x}(y) \in \mathbb{R}$ for its i th mean coordinate. Differentiating the quotient defining b_i gives

$$\partial_j b_i = \frac{1}{\varepsilon} \left(\int y_i y_j d\pi_{\varepsilon,x}(y) - b_i b_j \right), \quad \partial_i \log \rho_\varepsilon(x) = \frac{b_i - x_i}{\varepsilon}.$$

All integrals in these formulas are finite. Indeed, a polynomial times the Gaussian kernel is bounded in y , locally uniformly in x . Therefore,

$$\frac{a}{1+a\varepsilon} \text{Id} \preceq D^2V_\varepsilon(x) = \varepsilon^{-1} \text{Id} - \varepsilon^{-2} \text{Cov}(\pi_{\varepsilon,x}) \preceq \varepsilon^{-1} \text{Id}.$$

The lower bound is the one just proved; the upper bound follows from positivity of the covariance matrix.

Finally, couple all the variables as $X + \sqrt{\varepsilon}G$. For $0 < \varepsilon \leq 1$ and every $k > 0$,

$$|X + \sqrt{\varepsilon}G|^k \leq 2^{\max\{k-1,0\}}(|X|^k + |G|^k).$$

Finite moments and dominated convergence give both asserted limits.

For the normalized measures in Equation (F.1), the covariance is $(\text{Cov}(\mu) + \varepsilon \text{Id})/(1+\varepsilon) \preceq \text{Id}$. The chain rule gives

$$\frac{a(1+\varepsilon)}{1+a\varepsilon} \text{Id} \preceq D^2V_{\mu_\varepsilon} \preceq \frac{1+\varepsilon}{\varepsilon} \text{Id}.$$

The lower bound is at least $a\text{Id}$ when $a \leq 1$. The additional scale tends to one, so the weak and moment limits are preserved. $\square$

F.2 Regular Isotropic Approximation

Combining Gaussian regularization with a quadratic tilt and affine normalization gives the regular isotropic approximations needed in our main theorem. The next lemma also records convergence of every fixed moment.

Lemma F.2 (Regular Isotropic Approximation). *Let μ be an isotropic log-concave probability measure on $\mathbb{R}^n$. There are regular isotropic probability measures μ_j converging weakly to μ , with convergence of every fixed moment. More precisely, $\int p d\mu_j \rightarrow \int p d\mu$ for every polynomial $p : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\int |x|^k d\mu_j \rightarrow \int |x|^k d\mu$ for every real $k > 0$. Their lower Hessian bounds may tend to zero and their upper Hessian bounds may diverge.*

Proof. Gaussian regularization supplies a smooth upper Hessian bound. A quadratic tilt supplies a positive lower bound, and an affine normalization restores isotropy. Let $X \sim \mu$, let G be an independent standard Gaussian, and let ρ_ε and V_ε be as in Lemma F.1.

For integers $j \geq 1$, choose $\varepsilon_j = \delta_j = 1/j$, put $X_j = X + \sqrt{\varepsilon_j}G \in \mathbb{R}^n$, and define a probability measure ν_j on $\mathbb{R}^n$ by

$$d\nu_j(x) = Z_j^{-1} e^{-\delta_j |x|^2/2} \rho_{\varepsilon_j}(x) dx, \quad Z_j = \mathbb{E} e^{-\delta_j |X_j|^2/2}.$$

Here, $0 < Z_j \leq 1$. Its smooth potential $W_j = V_{\varepsilon_j} + \delta_j |x|^2/2 \in C^\infty(\mathbb{R}^n)$ satisfies

$$\delta_j \text{Id} \preceq D^2 W_j(x) \preceq (\delta_j + \varepsilon_j^{-1}) \text{Id}. \quad (\text{F.2})$$

We next check that neither the tilt nor the normalization changes the limit. In the coupling just chosen, $X_j \rightarrow X$ almost surely and $\delta_j |X_j|^2 \rightarrow 0$. Since $\varepsilon_j \leq 1$,

$$|X_j|^2 \leq 2|X|^2 + 2|G|^2.$$

The right-hand side is integrable. Dominated convergence first gives $Z_j \rightarrow 1$. For every bounded continuous $f : \mathbb{R}^n \rightarrow \mathbb{R}$, it also gives

$$\int f d\nu_j = Z_j^{-1} \mathbb{E} [f(X_j) e^{-\delta_j |X_j|^2/2}] \rightarrow \mathbb{E} f(X).$$

For any fixed $k > 0$,

$$|X_j|^k \leq 2^{\max\{k-1, 0\}} (|X|^k + |G|^k).$$

This bound is integrable, so the same argument proves convergence of every polynomial moment and of $\int |x|^k d\nu_j$. In particular, the mean $m_j \in \mathbb{R}^n$ and covariance $S_j \in \text{Sym}^2 \mathbb{R}^n$ of ν_j satisfy

$$m_j \rightarrow 0, \quad S_j \rightarrow \text{Id}.$$

Each S_j is positive definite. Zero variance in a nonzero direction would force ν_j to be supported on a hyperplane. This is impossible because its density is strictly positive everywhere.

Let μ_j be the image of ν_j under $x \mapsto S_j^{-1/2}(x - m_j)$. It is centered and has covariance Id . Its potential $\widetilde{W}_j \in C^\infty(\mathbb{R}^n)$, up to an additive constant, is

$$\widetilde{W}_j(y) = W_j(m_j + S_j^{1/2}y).$$

The chain rule and Equation (F.2) give

$$\delta_j \lambda_{\min}(S_j) \text{Id} \preceq D^2 \widetilde{W}_j(y) \preceq (\delta_j + \varepsilon_j^{-1}) \lambda_{\max}(S_j) \text{Id}.$$

Both constants are finite and positive, so μ_j is regular.

Finally, $S_j^{-1/2} \rightarrow \text{Id}$ and $m_j \rightarrow 0$. For bounded continuous f ,

$$\int f d\mu_j = Z_j^{-1} \mathbb{E} [f(S_j^{-1/2}(X_j - m_j)) e^{-\delta_j |X_j|^2/2}] \rightarrow \mathbb{E} f(X)$$

by dominated convergence in the same coupling. This proves weak convergence. Since $S_j^{-1/2} \rightarrow \text{Id}$ and $m_j \rightarrow 0$, there is a finite B such that $\|S_j^{-1/2}\|_{\text{op}} \leq B$ and $|m_j| \leq B$ for all j . For each fixed $k > 0$, therefore,

$$|S_j^{-1/2}(X_j - m_j)|^k \leq C_{k,B} (1 + |X|^k + |G|^k).$$

The integrable right-hand side also dominates every polynomial of degree at most k , up to a constant depending on its coefficients. Dominated convergence in the last displayed expectation proves all the asserted moment convergences. $\square$

F.3 Extension to Finite-Energy Functions

The weak-limit argument gives the inequality for smooth, compactly supported functions. We now extend it to a larger class of test functions. The final lemma shows that if the squared gradient is integrable, then the function is square-integrable too, and the same Poincaré bound applies.

Lemma F.3 (Finite-Energy Test Functions). *Let μ be an absolutely continuous probability measure on $\mathbb{R}^n$. Suppose that, for some $0 \leq K < \infty$,*

$$\mathrm{Var}_\mu u \leq K \int |\nabla u|^2 d\mu \quad (u \in C_c^\infty(\mathbb{R}^n)). \quad (\text{F.3})$$

Then every locally Lipschitz $f : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfying $\int |\nabla f|^2 d\mu < \infty$ belongs to $L^2(\mu)$ and satisfies

$$\mathrm{Var}_\mu f \leq K \int |\nabla f|^2 d\mu. \quad (\text{F.4})$$

If the density of μ is smooth and strictly positive, the same conclusion holds for every $f \in W_{\mathrm{loc}}^{1,2}(\mathbb{R}^n)$ whose weak gradient has finite $L^2(\mu)$ norm.

Proof. We first extend Equation (F.3) to compactly supported Lipschitz functions. Let $u : \mathbb{R}^n \rightarrow \mathbb{R}$ be such a function. Choose a nonnegative mollifier $\varphi_\varepsilon \in C_c^\infty(\mathbb{R}^n)$ supported in the ball of radius ε and having integral one. Set $u_\varepsilon = u * \varphi_\varepsilon$. Then $u_\varepsilon \in C_c^\infty(\mathbb{R}^n)$ and $\|u_\varepsilon - u\|_\infty \rightarrow 0$. If L is a Lipschitz constant for u , its weak gradient satisfies $|\nabla u| \leq L$ almost everywhere with respect to Lebesgue measure, and $\nabla u_\varepsilon = (\nabla u) * \varphi_\varepsilon$. Consequently $|\nabla u_\varepsilon| \leq L$, and the Lebesgue differentiation theorem gives $\nabla u_\varepsilon \rightarrow \nabla u$ at almost every point with respect to Lebesgue measure. Absolute continuity of μ gives the same convergence μ -almost everywhere. Dominated convergence therefore allows us to pass to the limit in Equation (F.3), proving that inequality for u .

For the second assertion, a compactly supported $W_{\mathrm{loc}}^{1,2}$ function can also be mollified in the $W^{1,2}$ norm. The weight and its reciprocal are bounded on a fixed enlarged compact set. Thus, this approximation converges in the weighted function and gradient norms as well.

Next let $h : \mathbb{R}^n \rightarrow \mathbb{R}$ be bounded and locally Lipschitz, with $\nabla h \in L^2(\mu; \mathbb{R}^n)$; under the additional density assumption it may instead be bounded and locally $W^{1,2}$. Choose $\chi \in C_c^\infty(\mathbb{R}^n)$ taking values in $[0, 1]$, equal to one on the unit ball and zero outside the ball of radius two. Set $\chi_R(x) = \chi(x/R)$ and $h_R = \chi_R h$. The compact-support case just proved applies to h_R . We have $h_R \rightarrow h$ in $L^2(\mu)$. Moreover,

$$\nabla h_R - \nabla h = (\chi_R - 1)\nabla h + h\nabla \chi_R.$$

The first term tends to zero in $L^2(\mu)$ by dominated convergence, whereas the second has $L^2(\mu)$ norm at most $\|h\|_\infty \|\nabla \chi\|_\infty / R$. Passing to the limit gives

$$\mathrm{Var}_\mu h \leq K \int |\nabla h|^2 d\mu. \quad (\text{F.5})$$

Finally, let f have finite energy $E = \int |\nabla f|^2 d\mu$, without assuming square integrability. For $N > 0$, define $f_N : \mathbb{R}^n \rightarrow \mathbb{R}$ by $f_N = \max\{-N, \min\{f, N\}\}$. These functions are bounded and belong to the same local function class as f . The truncation chain rule gives $|\nabla f_N| \leq |\nabla f|$ almost everywhere, so Equation (F.5) yields

$$\mathrm{Var}_\mu f_N \leq KE.$$

We next bound the means of the truncations using a set where the original function is bounded. Choose $a \in [0, \infty)$ such that the set $A = \{|f| \leq a\}$ has probability $p = \mu(A) \in (0, 1]$; such an a exists because f is finite almost everywhere. For $N \geq a$, put $m_N = \int f_N d\mu \in \mathbb{R}$. On A we have $f_N = f$ and $|f_N| \leq a$, and hence

$$p(|m_N| - a)_+^2 \leq \int_A |f_N - m_N|^2 d\mu \leq \text{Var}_\mu f_N \leq KE.$$

Here, $s_+ = \max\{s, 0\}$ denotes the positive part of a real number. Therefore,

$$\int f_N^2 d\mu = \text{Var}_\mu f_N + m_N^2 \leq KE + (a + \sqrt{KE/p})^2.$$

Fatou's lemma now implies $f \in L^2(\mu)$. Since $|f_N - f| \leq |f|$ and $f_N \rightarrow f$ pointwise, dominated convergence gives $f_N \rightarrow f$ in $L^2(\mu)$. Thus, $\text{Var}_\mu f = \lim_{N \rightarrow \infty} \text{Var}_\mu f_N \leq KE$, which is Equation (F.4). $\square$

G An Explicit Poincaré Constant

We record an admissible value of the universal Poincaré constant in Theorem 1.1. The coefficient induction uses the radius $R_* = 10^8$ specified in Table 1. Corollary 8.2 gives the Poincaré bound $1 + 2R_*^2$, and the approximation argument in Section 9 preserves that constant. Thus, one may take

$$K_P = 1 + 2 \cdot 10^{16} < 3 \cdot 10^{16}.$$